

Autodesk® Scaleform®

Scaleform CLIK

Scaleform CLIK

: Prasad Silva Matthew Doyle
: 2.0
: 2010 8 19

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Autodesk® Scaleform® 4.2

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1 . @ *

Scaleform® Common Lightweight Interface Kit (CLIK™)

CLIK User Guide

[Getting Started with CLIK](#)

2

[Getting Started with CLIK Buttons](#)

Scaleform CLIK

CLIK

1.1 . @ *

Scaleform CLIK ActionScript™ 2.0 (AS) (UI)

PC

Scaleform 4.1

(

UI
CPU

UI

)

CLIK 15
()

1.1.1 Scaleform CLIK - Ô •

e ¥ • 4 • ¥ z:

UI

Button	Slider
ButtonBar	StatusIndicator
CheckBox	ProgressBar
RadioButton	UILoader
Label	ScrollingList
TextInput	TileList
TextArea	DropDownMenu
ScrollIndicator	Dialog
ScrollBar	NumericStepper

a > k:

API

InputManager	Locale
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IdataProvider

EventDispatcher	Ilist
Delegate	IlistItemRenderer
Constraints	GameDataProvider

• } — T • ' g ¥ % • ‡ S V • :

Getting Started with CLIK
Getting Started with CLIK Buttons
CLIK API Reference
CLIK User Guide
CLIK Flash Samples
CLIK Video Tutorials

1.2 e ¥ • 4 • ¥ z * # %

Flash

Flash

gskinner.com

Scaleform

Scaleform CLIK

gskinner.com	Adobe
Flash Creative Suite® 4 (CS4)	
gskinner.com	Flash
gskinner.com	1
	http://gskinner.com/blog

CLIK

CLIK

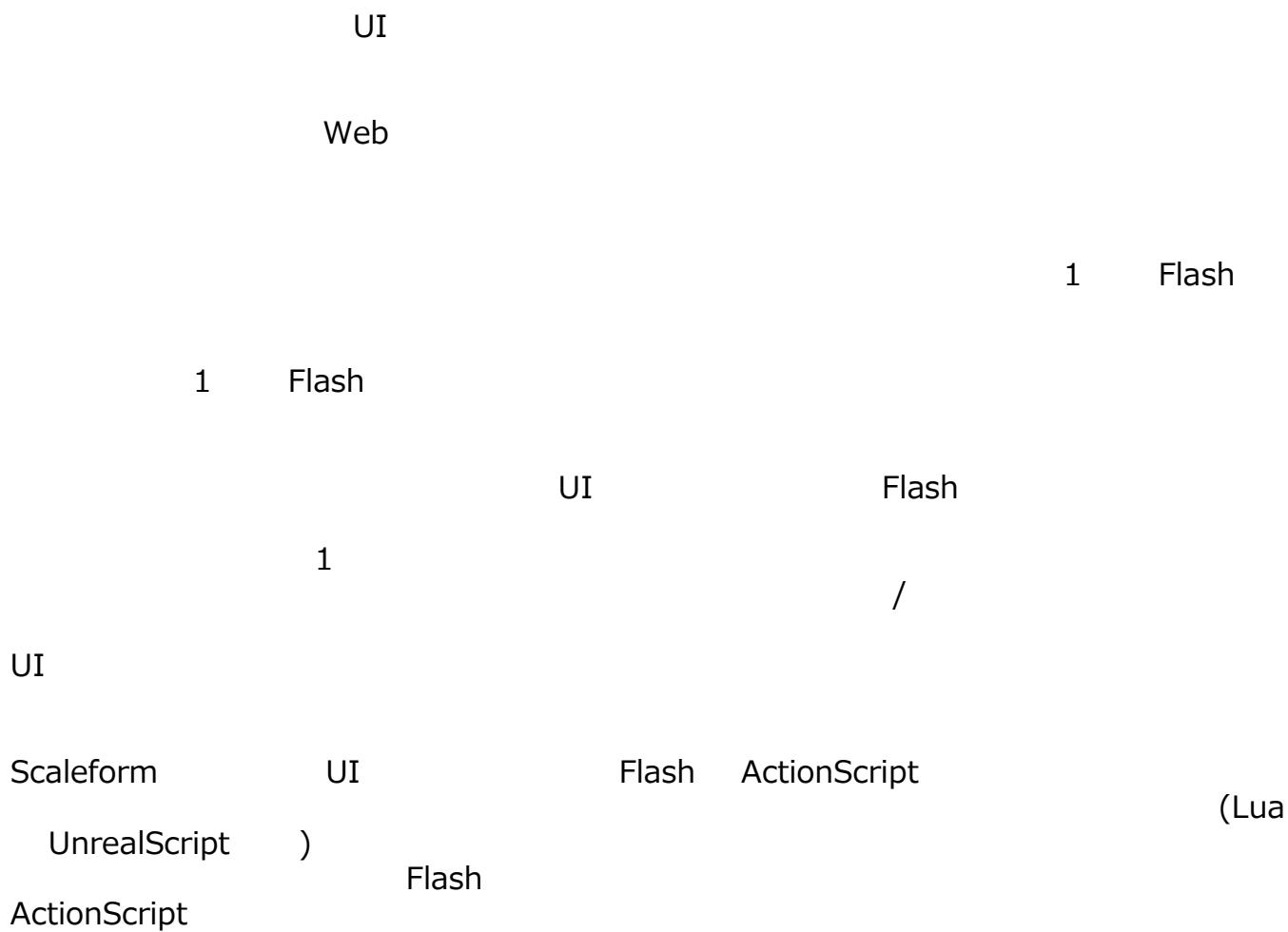
Flash Studio

:

C:\Program Files\Scaleform\GFx 4.2\Resources\AS2\CLIK\components\CLIK_Components.fla

1.3 UI * # % - ... Ä

UI



2 %o œ ... • { e ¥ • 4 • ¥ z

CLIK Scaleform CLIK UI CLIK

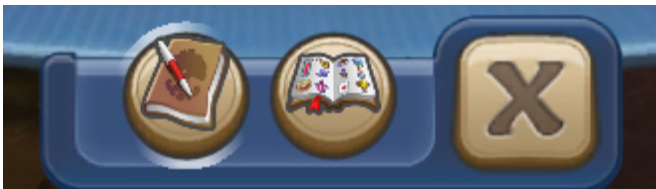
CLIK

UI

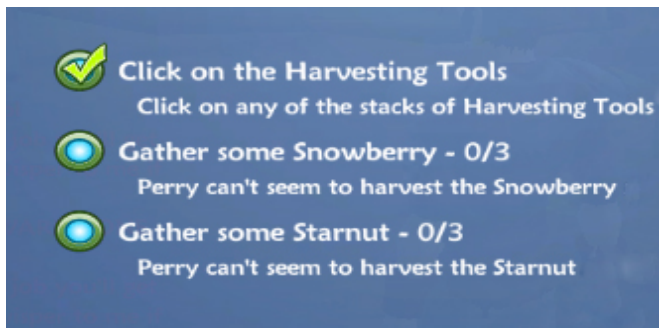
- ~ 4 h 4 - • • • :
- e ¥ • 4 • ¥ z - m u z T u ‰ : Flash
- k x 4 z : (
- ^ ' í š) ‰ Ÿ f x U :
Flash
- V < ¥ z : UI
- „ ¥ z ' e v :

2.1 f ® Ó) Ž q ¥ ' x _ k z - q V ‰

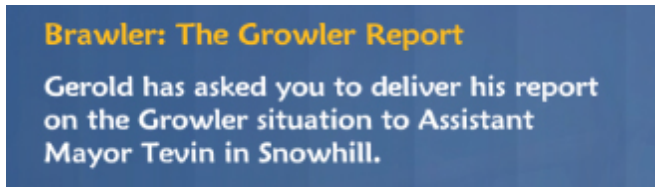
Button CheckBox Label TextInput TextArea
CheckBox Button Label
TextInput TextArea textField



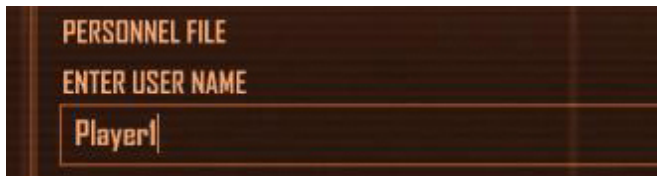
1: Free Realms –



2: *Free Realms* -

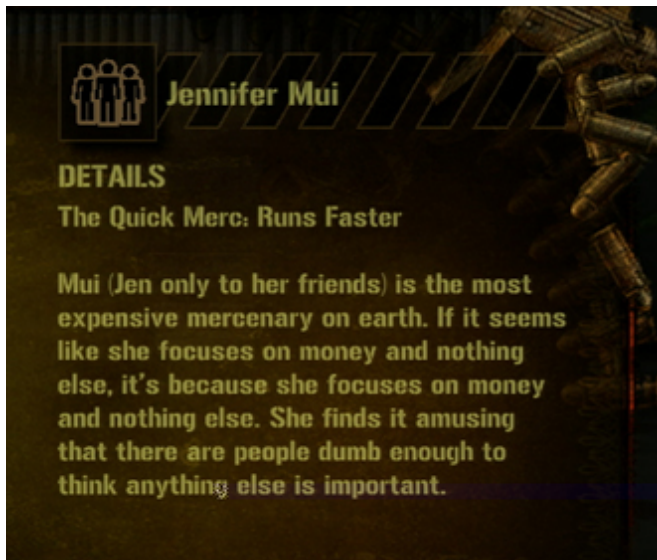


3: *Free Realms* -



4: *Crysis Warhead* -

TextInput



5: *Mercenaries 2* -

TextArea

2.1.1 Button



6:

Button CLIK

(gfx.controls.Button)

Button

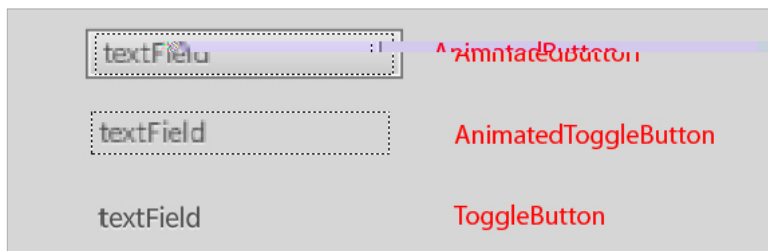
textField

ScrollBar

Button

Slider

Button



7: `AnimatedButton` `AnimatedToggleButton` `ToggleButton`

CLIK Button

Button.fla

ToggleButton AnimatedButton

AnimatedToggleButton

2.1.1.1 ~ 4 h 4 - ••

Button

CLIK

Tab

Shift+Tab

$$\begin{pmatrix} \cdot \\ \cdot \end{pmatrix}$$

)

Web
 Shift+Tab
 CLIK
 CLIK
 Flash
 Tab
 CLIK
 Button
 Button
 Enter
 Button
 /
 Xbox360
 PS3
 CLIK
 A
 UI
 X
 Enter

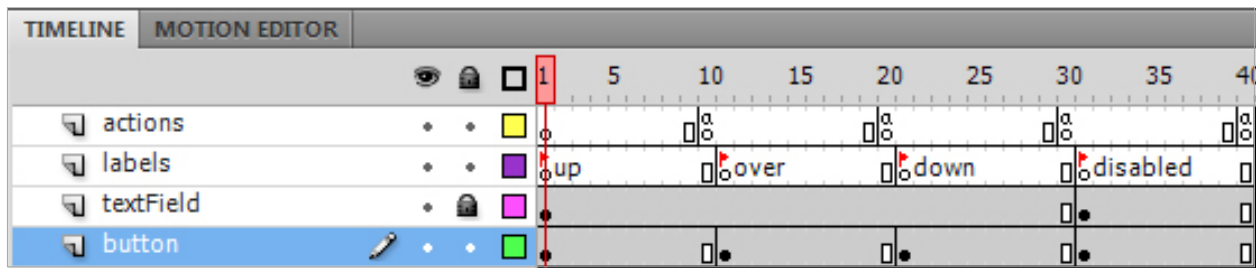
2.1.1.2 e ¥ • 4 • ¥ z - m u z T u %o

CLIK Button
 MovieClip
 • *textField* : () TextField
 • *focusIndicator* : () MovieClip
 MovieClip
 "hide" 2
over
 Button
 MovieClip "show"
 focusIndicator
 Button

2.1.1.3 k x 4 z

CLIK Button
 :
 • *up*
 •
over
 • *down*

- *disabled*



8: Button

Flash

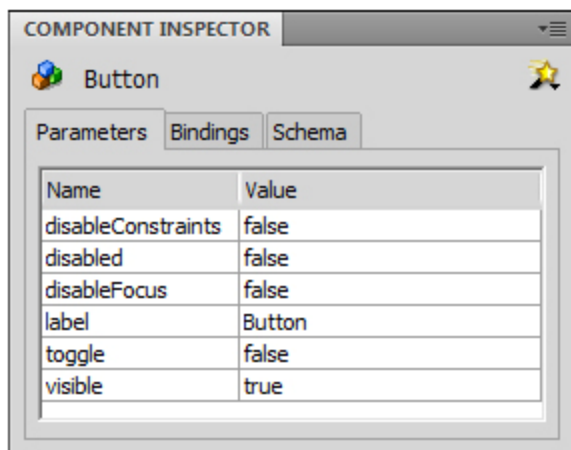
Button

Buttons

[Getting Started with CLIK](#)

2.1.1.4 ^ ' í š) % ¨ Ÿ f x U

Flash IDE [CS4 [Shift
+ F7 [AS



9: CS4 [Button

Button

:

disableConstraints Button

		constraints (	textField	)	constraints
		true			
		textField			
		textField			
		textField		/	
Disabled		true			
disableFocus					true
Label		Button			
Toggle		Button	(	)	true
			Button		
Visible		false			

Flash IDE SWF
 CLIK

2.1.1.5 V < ¥ z

CLIK

1 Object

- *type* :
- *target* :

Button

Show	visible	true
Hide	visible	false
focusIn		
focusOut		

Select	<code>selected</code> <code>selected : Button</code>	<code>true</code>
stateChange	<code>state : Button</code> <code>"up" "over" "down"</code> Getting Started with CLIK Buttons	
rollOver	<code>mouseIndex :</code> <code>(</code> <code>)</code>	<code>0 3</code>
rollOut	<code>mouseIndex :</code> <code>(</code> <code>)</code>	<code>0 3</code>
Press	<code>mouseIndex :</code> <code>(</code> <code>)</code>	<code>0 3</code>
doubleClick	<code>true</code> <code>mouseIndex :</code> <code>(</code> <code>)</code>	<code>doubleClickEnabled</code> <code>0 3</code>
Click	<code>mouseIndex :</code> <code>(</code> <code>)</code>	<code>0 3</code>
dragOver	<code>(</code> <code>mouseIndex :</code> <code>(</code> <code>)</code>	<code>)</code> <code>0 3</code>
dragOut	<code>(</code> <code>mouseIndex :</code> <code>(</code> <code>)</code>	<code>)</code> <code>0 3</code>
releaseOutside	<code>mouseIndex :</code> <code>(</code> <code>)</code>	<code>0 3</code>

ActionScript

:

```
myButton.addEventListener("click", this, "onButtonPress");
function onButtonPress(event:Object) {
    //
}
```

```
'myButton'           "click"
                      onButtonPress
Object                (      'event'      )
                      Object
```

2.1.1.6 „¥ z ' e v

Button :

```
import gfx.controls.Button;
attachMovie("Button", "btnInstanceName", someDepth, {_x:33, _y:262, ...});
```

CLIK Button

toggle

:

```
btn.toggle = true;
```

:

```
btn.doubleClickEnabled = true;
btn.addEventListener("doubleClick", this, "onDoubleClick");
function onDoubleClick(e:Object):Void {
    //
    !
}
```

click :

```
btn.autoRepeat = true;
```

2.1.2 CheckBox



10: CheckBox

CheckBox (gfx.controls.CheckBox)
Button

CheckBoxes true/false ()
ToggleButton

toggle

2.1.2.1 ~ 4 h 4 - • •

CheckBox

CheckBox Button

2.1.2.2 e ¥ • 4 • ¥ z - m u z T u %o

CLIK CheckBox

MovieClip

:

- *textField* : () TextField
 - *focusIndicator* : () MovieClip
- MovieClip "hide" 2

MovieClip "show"

2.1.2.3 k x 4 z

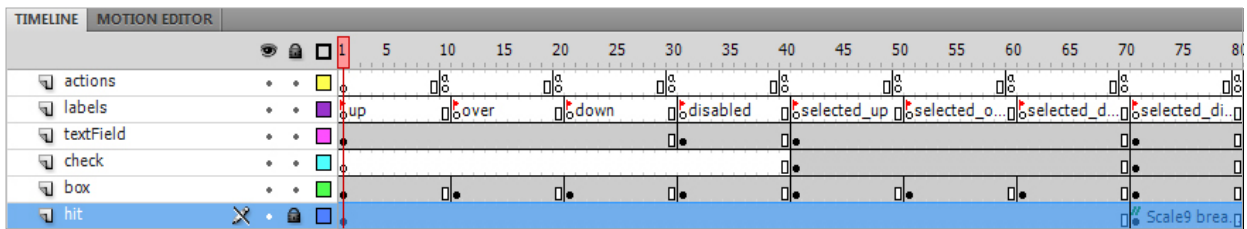
toggle

CheckBox

:

- *up*

- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*



11: CheckBox

CheckBox

Button
CheckBox

[Getting Started with CLIK Buttons](#)

2.1.2.4 ^ ' í š) % ¨ Ÿ f x U

CheckBox Button

toggle disableFocus

Button

Data	CheckBox
disableConstraints	Button
	constraints (true)
	textField
	textField
	textField /
disabled	true
Label	Button

selected	true	CheckBox	(
Visible	false		

2.1.2.5 V < ¥ z

1 Object

- *type* :
- *target* :

CheckBox

Show	visible	true
Hide	visible	false
focusIn		
focusOut		
Select	selected	
	<i>selected</i> : Button	selected
stateChange	<i>state</i> : Button "up" "over" "down" Getting Started with CLIK Buttons	
rollOver	<i>mouseIndex</i> :	
	(	0 3
	)	
rollOut	<i>mouseIndex</i> :	
	(	0 3
	)	
Press	<i>mouseIndex</i> :	
	(	0 3
	)	
doubleClick	true <i>mouseIndex</i> :	<i>doubleClickEnabled</i>
	(	0 3
	)	
Click	<i>mouseIndex</i> :	

	(	0	3
dragOver	(		)
	<i>mouseIndex :</i>	(	
	)	0	3
dragOut	(		)
	<i>mouseIndex :</i>	(	
	)	0	3
releaseOutside	(		)
	<i>mouseIndex :</i>	(	
	)	0	3

CheckBox

```
myCheckBox.addEventListener("select", this, "onCheckBoxToggle");
function onCheckBoxToggle(event:Object) {
    //
}
```

2.1.3 Label



12: Label

```
CLIK Label (gfx.controls.Label)
MovieClip
Label textField
Label textField
Label
textField
Label
```

textField	Label
-----------	-------

2.1.3.1 ~ 4 h 4 - ••

Label

2.1.3.2 e ¥ • 4 • ¥ z - m u z T u %

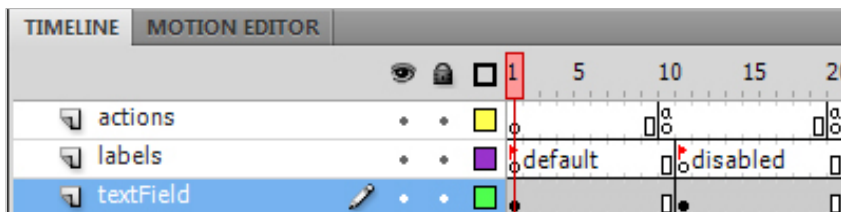
CLIK Label MovieClip :

- textField : TextField Label

2.1.3.3 k x 4 z

CLIK Label	disabled	2
:		

- *default*
- *disabled*



13: Label

2.1.3.4 ^ ' í š) % ¨ Ÿ f x U

Label :

Text	
Visible	false
disabled	true

2.1.3.5 V < ¥ z

1 Object

- *type* :
- *target* :

Label

Show	visible	true
Hide	visible	false
stateChange	Label <i>state</i> : "default"	Label "disabled"

Label :

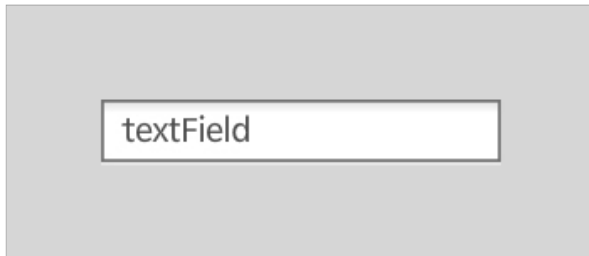
```
myLabel.addEventListener("stateChange", this, "onStateChange");
function onStateChange(event:Object) {
    if (event.state == "default") {
        //
    }
}
```

2.1.3.6 „ ¥ z ' e v

Label HTML textField
HTML Flash 8 :

```
lbl.htmlText = "<b>foo</b>bar";
```

2.1.4 TextInput



14: TextInput

TextInput (gfx.controls.TextInput)
textField

textField
textField

Label

HTML

TextInput textField

TextInput

Label
textField
UI

TextInput

Label
TextInput PC
textField

TextInput focused disabled
textField

TextInput
textField

AS2

TextInput
TextInput

textField

Flash Studio
TextInput

TextInput

HTML

textField

(Shift+)
(Shift+Insert)

(Shift+Delete)

(Ctrl+Insert)

TextInput

actAsButton inspectable

“default” “over” rollOver rollOut “over” “out”
 actAsButton
 TextInput CLIK

Text String
 0xAAAAAA
 TextFormat Text

2.1.4.1 ~ 4 h 4 - • •

TextInput
 textField
 textField

2.1.4.2 e ¥ • 4 • ¥ z - m u z T u %o

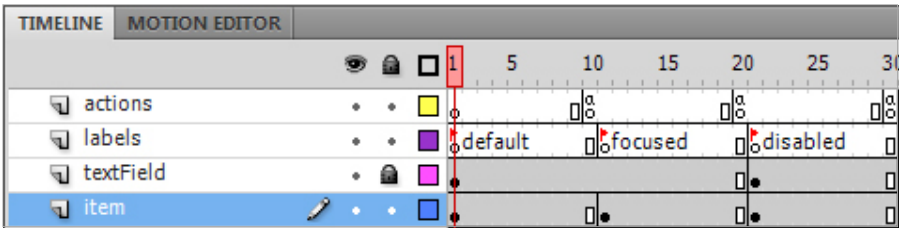
CLIK TextInput MovieClip :

- textFie ld : textField

2.1.4.3 k x 4 z

CLIK TextInput : focused disabled 3

- *default*
- *focused* textField
- *disabled*



15: TextInput

2.1.4.4 ^ ' í š) % ¨ Ÿ f x U

TextInput :

Text	textField				
Visible	false				
disabled	true				
editable	false		TextInput		
maxChars			textField		
password	true			'*'	textField
		textField			
	text				
defaultText	textField				
	defaultTextFormat				
actAsButton	true	TextInput			Button
		rollOver	rollOut		
		TextInput			

2.1.4.5 V < ¥ z

1 Object

- type :
- target :

TextInput

Show	visible	true
Hide	visible	false

focusIn	
focusOut	
textChange	textField
rollOver	actAsButton
	<i>mouseIndex:</i>
	0 3
rollOut	actAsButton
	<i>mouseIndex:</i>
	0 3

textField :

```
myTextInput.addEventListener("textChange", this, "onTextChange");
function onTextChange(event:Object) {
    //
}
```

2.1.4.6 „¥ z ' e v

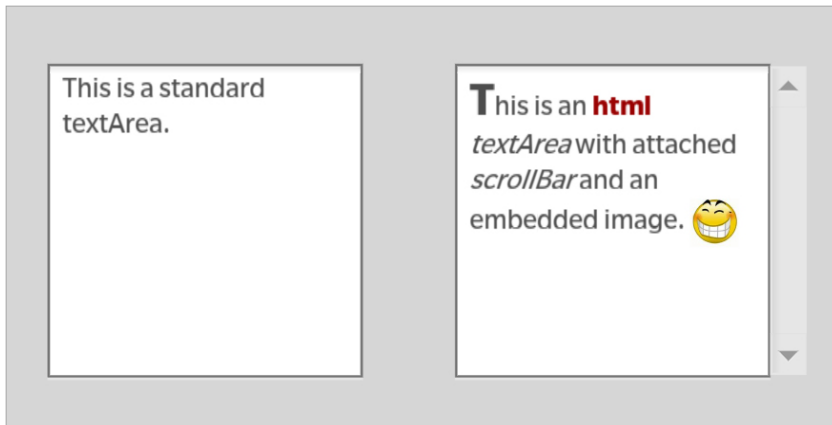
:

```
_global.gfxExtensions = true;
textInput.textField.noAutoSelection = true;
```

TextInput HTML HTML Flash 8 textField :

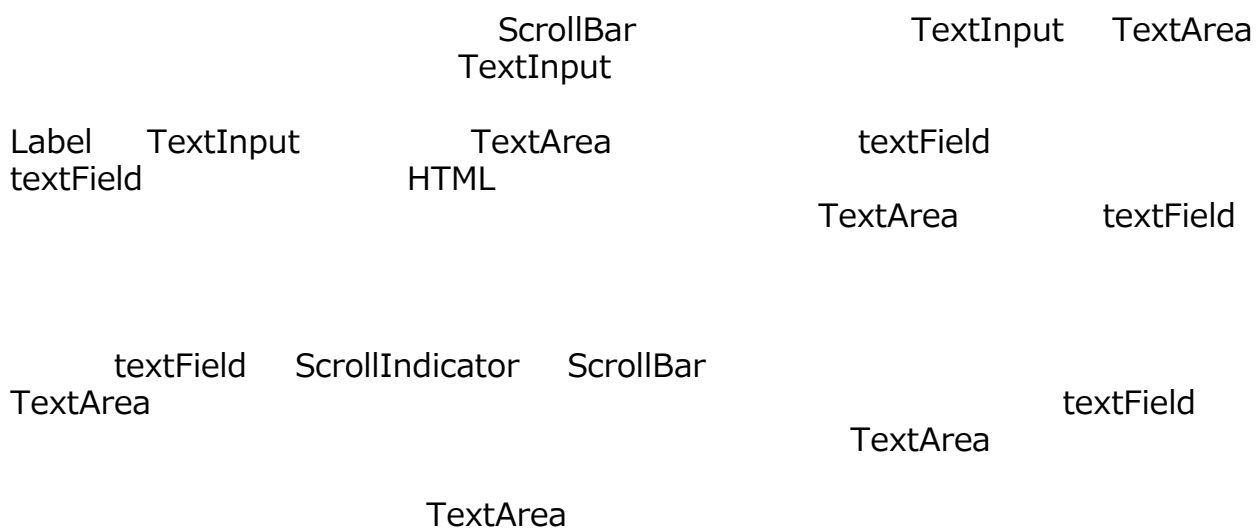
```
textInput.htmlText = "<b>foo</b>bar";
```

2.1.5 TextArea



16: TextArea

TextArea (gfx.controls.TextArea) CLIK TextInput



2.1.5.1 ~ 4 h 4 - ••

TextArea
textField

TextArea :

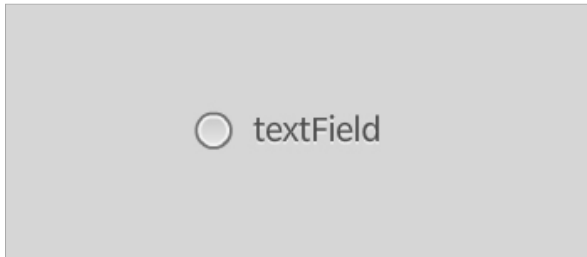
```
myTextArea.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
    //
}
```

2.1.5.6 „¥ z ' e v

:

```
_global.gfxExtensions = true;
```

2.2.1 RadioButton



19: RadioButton

RadioButton (gfx.controls.RadioButton) 1
 RadioButton 1

CLIK RadioButton CheckBox
 RadioButton
 ButtonGroup ()
 RadioButton toggle
 () ButtonGroup

2.2.1.1 ~ 4 h 4 - ••

RadioButton
 RadioButton
 RadioButton
 ButtonGroup
 RadioButton
 Button

2.2.1.2 e ¥ • 4 • ¥ z - m u z T u %

CLIK RadioButton MovieClip
 :

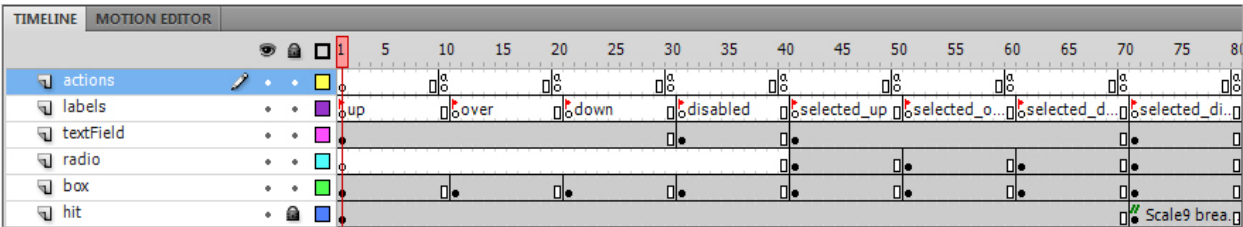
- textField : () TextField
- *focusIndicator* : () MovieClip

MovieClip "hide" 2 MovieClip "show"

2.2.1.3 k x 4 z

RadioButton
CheckBox :

- up
- over
- down
- disabled
- selected_up
- selected_over
- selected_down
- selected_disabled



20: RadioButton

RadioButton Button
RadioButton

[Getting Started with CLIK Buttons](#)

2.2.1.4 ^ ' í š) % ¨ Ÿ f x U

RadioButton Button
Button toggle disableFocus

Label	Button
Visible	false
disabled	true
disableConstraints	Button
	textField
	constraints ()
	true constraints

	textField		Button
	textField	Button	/
selected	true	RadioButton	(
Data	RadioButton		)
group	ButtonGroup	RadioButton	RadioButton
		ButtonGroup	RadioButton
		ButtonGroup	RadioButton _root
			_root
	RadioButton	1	

2.2.1.5 V < ¥ z

1 Object

- type :
- target :

RadioButton

Show	visible	true
Hide	visible	false
focusIn		
focusOut		
select	selected	selected
	<i>selected</i> : Button	selected
stateChange	<i>state</i> : Button	
	"up" "over" "down"	
	Getting Started with CLIK Buttons	
rollOver	<i>mouseIndex</i> :	

	<code>(</code>	<code>0</code>	<code>3</code>
rollOut	<code>)</code>		
	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
press	<code>)</code>	<code>3</code>	
	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
doubleClick	<code>)</code>	<code>3</code>	<code>doubleClickEnabled</code>
	<code>true</code>		
	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
click	<code>)</code>	<code>3</code>	
	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
dragOver	<code>)</code>	<code>3</code>	<code>)</code>
	<code>mouseIndex :</code>	<code>(</code>	
	<code>)</code>	<code>0</code>	<code>3</code>
dragOut	<code>(</code>		<code>)</code>
	<code>mouseIndex :</code>	<code>(</code>	
	<code>)</code>	<code>0</code>	<code>3</code>
releaseOutside			
	<code>mouseIndex :</code>	<code>(</code>	
	<code>)</code>	<code>0</code>	<code>3</code>

RadioButton :

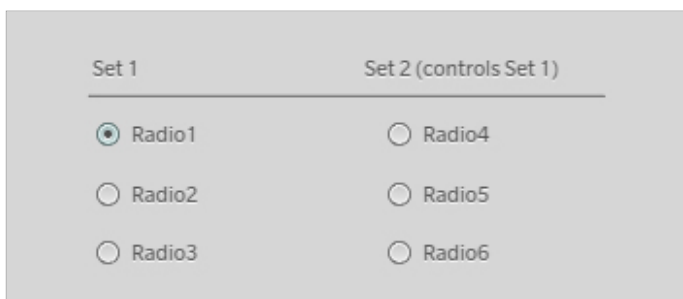
```
myRadio.addEventListener("select", this, "onRadioToggle");
function onRadioToggle(event:Object) {
    //
}
```

2.2.1.6 „¥ z ' e v

RadioButton :

```
import gfx.controls.ButtonGroup;
var myGroup:ButtonGroup = new ButtonGroup("groupName");
radio1.group = myGroup;
radio2.group = myGroup;
someMovie.radio1.group = myGroup;
someMovie.radio2.group = myGroup;
```

2.2.2 ButtonGroup



21: ButtonGroup

CLIK ButtonGroup (gfx.controls.ButtonGroup)

1

Button (CheckBox RadioButton)
ButtonGroup

ButtonGroup

CLIK

2.2.2.1 ~ 4 h 4 - ••

ButtonGroup
RadioButton

2.2.2.2 e ¥ • 4 • ¥ z - m u z T u %

CLIK ButtonGroup MovieClip

2.2.2.3 k x 4 z

ButtonGroup

2.2.2.4 ^ ' í š) % ¨ Ÿ f x U

ButtonGroup

2.2.2.5 V < ¥ z

1 Object

- type :
- target :

ButtonGroup

change	• <i>item</i> : • <i>data</i> :	CLIK Button AS2 Object
itemClick	• <i>item</i> :	CLIK Button

ButtonGroup :

```
myGroup.addEventListener("change", this, "onNewSelection");
function onNewSelection(event:Object) {
    if (event.item == myRadio) {
        //
    }
}
```

2.2.2.6 „ ¥ z ' e v

RadioButton :

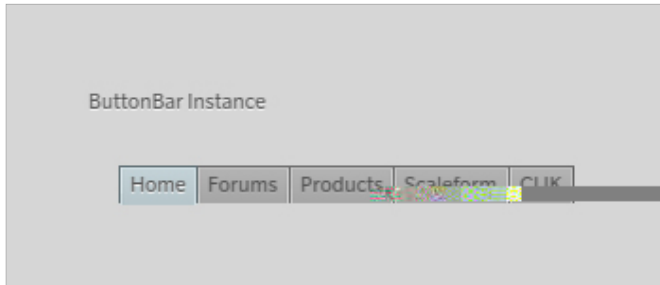
```
var selectedRadio:Button = group.selectedButton;
```

RadioButton
:

ButtonGroup

```
radioBtn1.group.addEventListener("itemClick", this, "onClick");
```

2.2.3 ButtonBar



22: ButtonBar

ButtonBar (gfx.controls.ButtonBar) ButtonGroup
dataProvider
(dataProvider
) ButtonBar
dataProvider
Button
UI
:

```
buttonBar.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

2.2.3.1 ~ 4 h 4 - ••

ButtonBar ButtonGroup
ButtonBar Button

2.2.3.2 e ¥ • 4 • ¥ z - m u z T u %

CLIK ButtonBar MovieClip

2.2.3.3 k x 4 z

CLIK ButtonBar Button

2.2.3.4 ^ ' í š) % ¨ Ÿ f x U

ButtonBar (Flash IDE

ButtonBar) Button

visible	false	ButtonBar	
disabled	true	ButtonBar	
itemRenderer	Button	ID	ButtonBar
direction	Button	horizontal	Button
		vertical	
spacing	Button		
	(direction	)	
autoSize	true	Button	
buttonWidth	Button		autoSize
	true		

2.2.3.5 V < ¥ z

1 Object

- type :
- target :

ButtonBar

show	visible	true
hide	visible	false
focusIn		
focusOut		
change	• index : ButtonBar	0
	1	

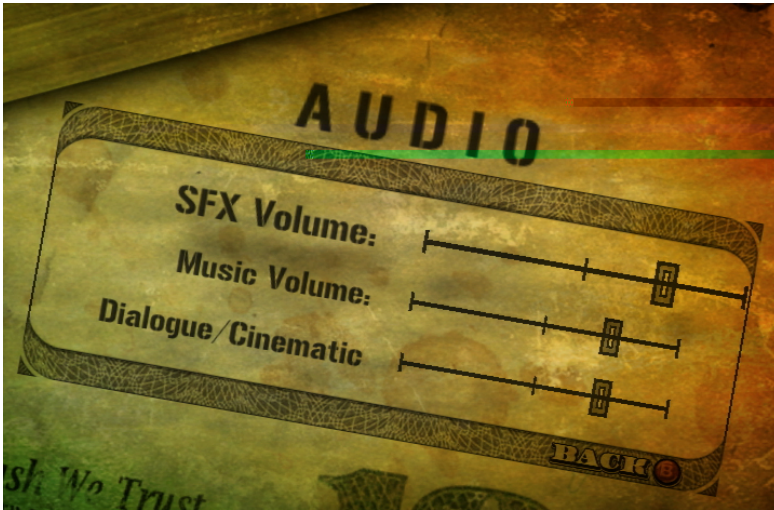
	• <i>item</i> : Button CLIK Button • <i>item</i> : dataProvider AS2 Object
	• <i>data</i> : dataProvider AS2 • Object
itemClick	• <i>index</i> : Button ButtonBar • <i>item</i> : dataProvider AS2 Object • <i>data</i> : dataProvider AS2 • <i>mouseIndex</i> : (0 3)

ButtonBar Button
:

```
myBar.addEventListener("itemClick", this, "onItemClick");
function onItemClick(event:Object) {
    processData(event.data);
    //
}
```

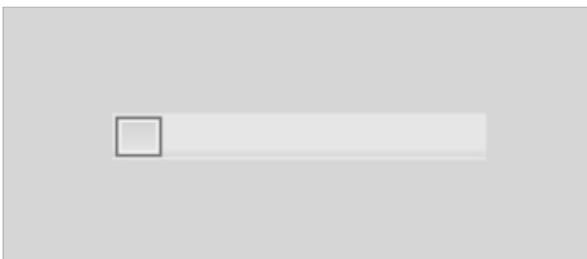

2.3 k a Ÿ 4 • q V ‰

ScrollIndicator ScrollBar Slider
 ScrollIndicator
) ScrollBar 4 ScrollBar
 Slider ScrollBar
 Slider ScrollBar



23: Mercenaries 2 -

2.3.1 ScrollIndicator



24: ScrollIndicator

CLIK ScrollIndicator (gfx.controls.ScrollIndicator) textField
 ScrollIndicator

textField
TextArea ScrollIndicator ScrollBar ()
ID scrollbar

2.3.1.1 ~ 4 h 4 - • •

ScrollIndicator

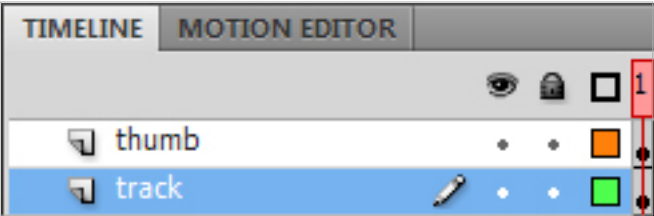
2.3.1.2 e ¥ • 4 • ¥ z - m u z T u %

CLIK ScrollIndicator MovieClip :

- *thumb* : CLIK Button
- *track* : MovieClip

2.3.1.3 k x 4 z

ScrollIndicator thumb
track Button



25: ScrollIndicator

2.3.1.4 ^ ' í š) % Ÿ f x U

ScrollIndicator :

scrollTarget	TextArea	textField	textField
ScrollIndicator			
visible	false		
disabled	true		
offsetTop	thumb		

offsetBottom	thumb
--------------	-------

2.3.1.5 V < ¥ z

1 Object

- *type* :
- *target* :

ScrollIndicator

show	visible	true
hide	visible	false
scroll	• <i>position</i> :	

:

```
mySI.addListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
    trace("scroll position: " + event.position);
    //
}
```

2.3.1.6 „¥ z ' e v

:

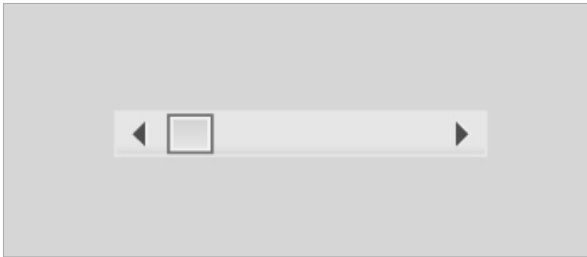
```
scrollInd.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollInd.positon = currPos;
```

_rotation
ScrollIndicator

:

```
scrollInd.direction = "horizontal";
```

2.3.2 ScrollBar



26: ScrollBar

CLIK ScrollBar (gfx.controls.ScrollBar)

ScrollIndicator

2.3.2.1 ~ 4 h 4 - • •

ScrollBar

ScrollBar

ScrollBar

2

1

1

trackMode inspectable
) trackMode

(
Shift

2.3.2.2 e ¥ • 4 • ¥ z - m u z T u %

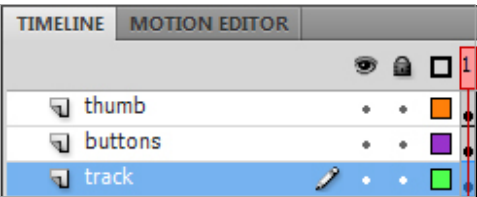
CLIK ScrollBar

MovieClip

:

- *upArrow* : CLIK Button Button
- *downArrow* : CLIK Button Button
- *thumb* : CLIK Button
- *track* : CLIK Button

2.3.2.3 k x 4 z



27: ScrollBar

ScrollIndicator ScrollBar
thumb up down track Button

2.3.2.4 ^ ' í š) % ¨ Ÿ f x U

ScrollBar ScrollIndicator 1 :

scrollTarget	TextArea	textField	textField
ScrollIndicator			
trackMode	scrollPage scrollToCursor		
visible	false		
disabled	true		
offsetTop	thumb		
offsetBottom	thumb		

2.3.2.5 V < ¥ z

1 Object

- type :
- target :

ScrollBar

show	visible	true
hide	visible	false

scroll

- *position:*

:

```
mySB.addEventListener("scroll", this, "onListScroll");
function onListScroll(event:Object) {
    trace("scroll position: " + event.position);
    //
}
```

2.3.2.6 „¥ z' e v

:

```
scrollBar.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollBar.positon = currPos;
```

_rotation
ScrollBar

:

```
scrollBar.direction = "horizontal";
```

scrollPage
1 :

```
scrollBar.trackScrollPageSize = pageSize;
```

2.3.3 Slider



28: Slider

Slider (gfx.controls.Slider)

2.3.3.1 ~ 4 h 4 - • •

Slider

Slider

Home

End

2.3.3.2 e ¥ • 4 • ¥ z - m u z T u %

CLIK Slider

MovieClip

:

- *thumb* : CLIK Button
- *track* : CLIK Button

2.3.3.3 k x 4 z

ScrollIndicator ScrollBar Slider
thumb track Button

2.3.3.4 ^ ' í š) % Ÿ f x U

Slider :

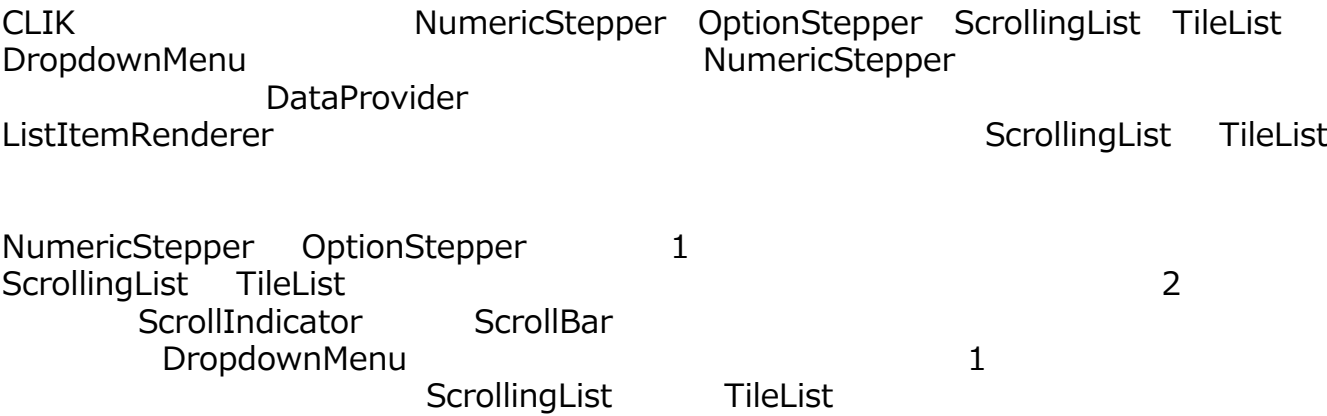
visible	false
disabled	true
value	Slider

Slider

Show	visible	true
Hide	visible	false
focusIn		
focusOut		
change	Slider	

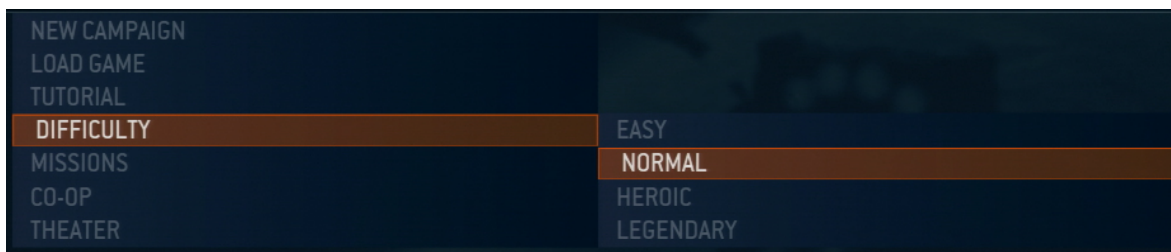
```
Slider :
mySlider.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("slider value: " + event.target.value);
    //
}
```

2.4 æ k z q V %

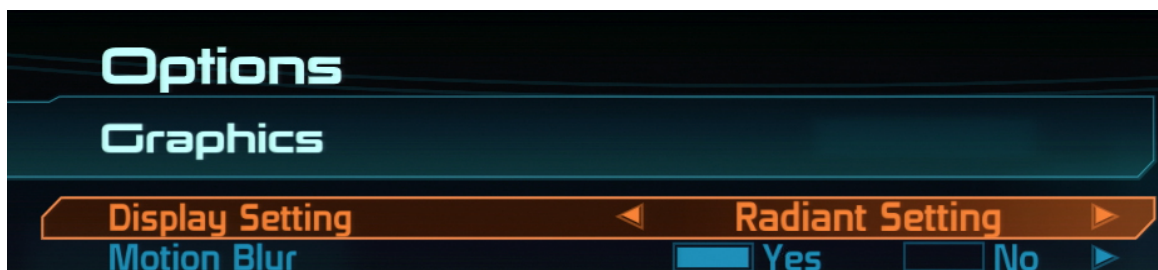




30: Mercenaries 2 -



31: Halo Wars -



32: Mass Effect -

2.4.1 NumericStepper



33: NumericStepper

NumericStepper (gfx.controls.NumericStepper) 1

2.4.1.1 ~ 4 h 4 - ••

NumericStepper 2

Home End

2.4.1.2 e ¥ • 4 • ¥ z - m u z T u %o

NumericStepper MovieClip :

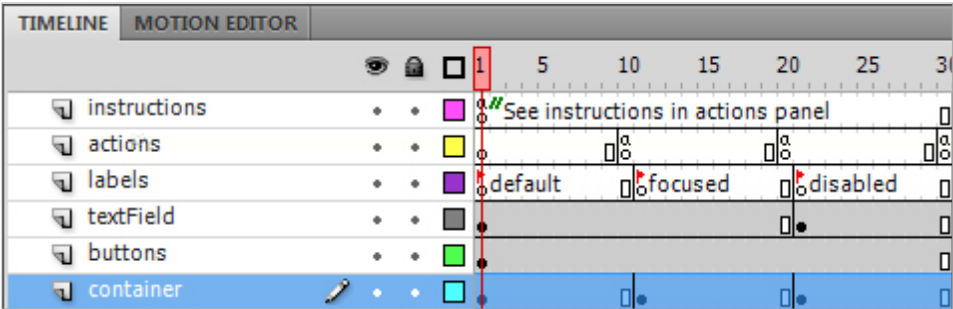
- *textField* : TextField
- *nextBtn* : CLIK Button
- *prevBtn* : CLIK Button

2.4.1.3 k x 4 z

NumericStepper : focused disabled 3

- *default*
- *focused* textField

- disabled



34: NumericStepper

2.4.1.4

NumericStepper

MovieClip

visible	false
disabled	true
value	NumericStepper
minimum	NumericStepper
maximum	NumericStepper

2.4.1.5

1 Object

- type :
- targe :

NumericStepper

show	visible	true
hide	visible	false
change	NumericStepper	
stateChange	NumericStepper	focused disabled
	state :	"default" "focused" "disabled"

NumericStepper :

```
myNS.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("ns value: " + event.target.value);
    //
}
```

2.4.1.6 „¥ z ' e v

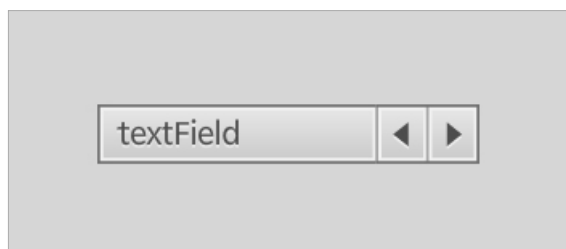
:

```
ns.stepSize = 0.5;
```

:

```
ns.labelFunction = function(value:Number) {
    switch(value) {
        case 1:
            return value + "st";
        default:
            return value + "th";
    }
}
```

2.4.2 OptionStepper



35: OptionStepper

NumericStepper OptionStepper (gfx.controls.OptionStepper) 1
dataProvider

OptionStepper selectedIndex

dataProvider dataProvider 0- :

```
optionStepper.dataProvider = ["item1", "item2", "item3", "item4"];
```

2.4.2.1 ~ 4 h 4 - • •

NumericStepper OptionStepper 2

Home End dataProvider

2.4.2.2 e ¥ • 4 • ¥ z - m u z T u %

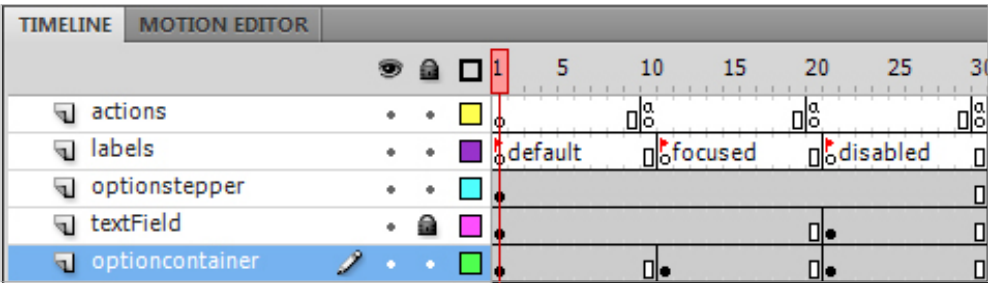
NumericStepper MovieClip :

- *textField* : TextField
- *nextBtn* : CLIK Button dataProvider
- *prevBtn* : CLIK Button dataProvider

2.4.2.3 k x 4 z

OptionStepper : focused disabled 3

- default
- focused textField
- disabled



36: OptionStepper

2.4.2.4 ^ ' í š) % º Ÿ f x U

OptionStepper : MovieClip

visible	false
disabled	true

2.4.2.5 V < ¥ z

1 Object

- type :
- target :

OptionStepper

show	visible	true
hide	visible	false

change	OptionStepper		
stateChange	OptionStepper	focused	disabled
	• state :		
	"default"	"focused"	"disabled"

OptionStepper :

```
myOS.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("os value: " + event.target.selectedItem);
    //
}
```

2.4.3 ListItemRenderer



37: ListItemRenderer

ListItemRenderer (gfx.controls.ListItemRenderer) CLIK Button

ScrollingList TileList DropdownMenu

2.4.3.1 ~ 4 h 4 - ••

ListItemRenderer Button
Button

ListItemRenderer

ListItemRenderer

2.4.3.2 e ¥ • 4 • ¥ z - m u z T u %

CLIK ListItemRenderer

MovieClip

:

- *textField* : () TextField
- *focusIndicator* : () MovieClip
- *hide* : MovieClip "show"

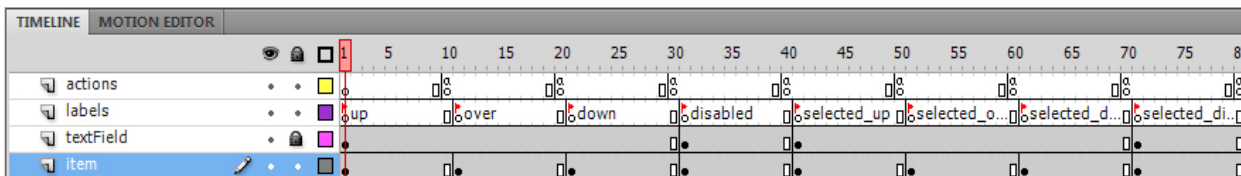
2.4.3.3 k x 4 z

ListItemRenderer

selected

:

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*



38: ListItemRenderer

ListItemRenderer

Button

ListItemRenderer

[Getting Started with CLIK Buttons](#)

$$\wedge \, ' \, \acute{\text{í}} \, \check{\text{š}} \,) \, \%_{00} \, \ddot{\text{Y}} \, f \, x \, \text{U}$$

ListItemRenderer

Button

label	ListItemRenderer
visible	false
disabled	true

2.4.3.5 $V < \neq Z$

1 Object

- *type* :
- *target* :

ListItemRenderer

Button

show	visible	true
hide	visible	false
focusIn		
focusOut		
select	selected	
	• <i>selected</i> : Button	selected
stateChange		
	• <i>state</i> : Button	
	"up" "over" "down"	
		Getting Started with CLIK Buttons
rollOver		
	• <i>mouseIndex</i> :	
		(0 3
rollOut		
	• <i>mouseIndex</i> :	
		(0 3
press		
	• <i>mouseIndex</i> :	
		(

doubleClick	• true • <i>mouseIndex</i> :	• 0 • 3	<i>doubleClickEnabled</i>
click	• <i>mouseIndex</i> :	• 0 • 3	
dragOver	• <i>mouseIndex</i> :	• 0 • 3	
dragOut	• <i>mouseIndex</i> :	• 0 • 3	
releaseOutside	• <i>mouseIndex</i> :	• 0 • 3	

2.4.4 ScrollingList



39: ScrollingList

ScrollingList (gfx.controls.ScrollingList)

ScrollIndicator ScrollBar

dataProvider : dataProvider

```
scrollingList.dataProvider = ["item1", "item2", "item3", "item4"];
```

```
ScrollingList
    itemRenderer
    ListItemRenderer
        FLA
    [
    ]
```

2.4.4.1 ~ 4 h 4 - ••

```

ScrollingList
    ScrollBar
        1
    ScrollingList
        wrapping
        ScrollingList
            wrapping
            "normal"
        wrapping
        "wrap"
        wrapping
        "stick"
        Page Up
        Page Down
        End
        Home
    ScrollingList
        ScrollBar
        ScrollBar
        ScrollBar
    D
        CLIK
        UI
```

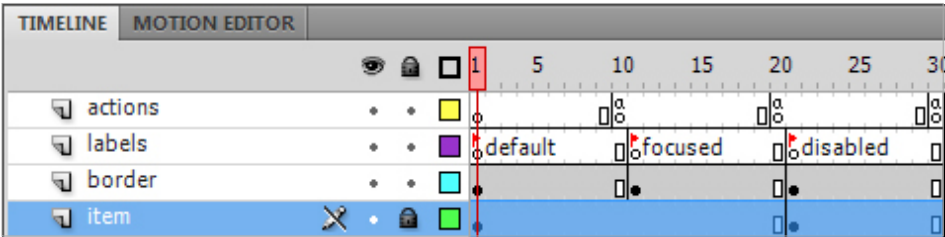
2.4.4.2 e ¥ • 4 • ¥ z - m u z T u %o

```
ScrollingList
    ScrollingList
```

2.4.4.3 k x 4 z

ScrollingList : focused disabled 3

- default
- focused
- disabled



40: ScrollingList

2.4.4.4 ^ ' í š) % º Ÿ f x U

ScrollingList : MovieClip

visible	false
disabled	true
itemRenderer	ListItemRenderer rendererInstanceName
renderInstanceName	ScrollingList
scrollBar	ScrollBar
	ScrollingList

ScrollingList		
margin	<i>rendererInstanceName</i>	
rowHeight	<i>rendererInstanceName</i>	
paddingTop	rendererInstanceName	
padding Bottom	rendererInstanceName	
paddingLeft	rendererInstanceName	
paddingRight	rendererInstanceName	
thumbOffsetTop		
thumbOffsetBottom		
thumbSizeFactor	1.0	1.0

2.4.4.5 V < ¥ z

1 Object

- *type* :
- *target* :

ScrollingList

show	visible	true
hide	visible	false
focusIn		
focusOut		
change	<i>index</i> : 0 1	
itemPress	<i>renderer</i> : CLIK Button <i>item</i> : dataProvider AS2 Object <i>index</i> : dataProvider 0 1 <i>mouseIndex</i> : (0 3)	
itemClick	<i>renderer</i> : CLIK Button <i>item</i> : dataProvider AS2 Object <i>index</i> : dataProvider 0 1 <i>mouseIndex</i> : (0 3)	
itemDoubleClick	<i>renderer</i> : CLIK Button <i>item</i> : dataProvider AS2 Object <i>index</i> : dataProvider 0 1 <i>mouseIndex</i> : (	

	)	0	3
itemRollOver			
• <i>renderer</i> :	Button		CLIK
• <i>item</i> :	dataProvider	AS2	Object
• <i>index</i> :	dataProvider	0	1
• <i>mouseIndex</i> :	(	0	3
	)		
itemRollOut			
• <i>renderer</i> :			CLIK Button
• <i>item</i> :	dataProvider	AS2	Object
• <i>index</i> :	dataProvider	0	1
• <i>mouseIndex</i> :	(	0	3
	)		

:

```
myList.addEventListener("itemClick", this, "onItemClicked");
function onItemClicked(event:Object) {
    trace("list item was clicked: " + event.renderer);
    //
}
```

2.4.4.6 „¥ z' e v

1 :

```
//
// ScrollingList 'label' :
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

// label
//
```



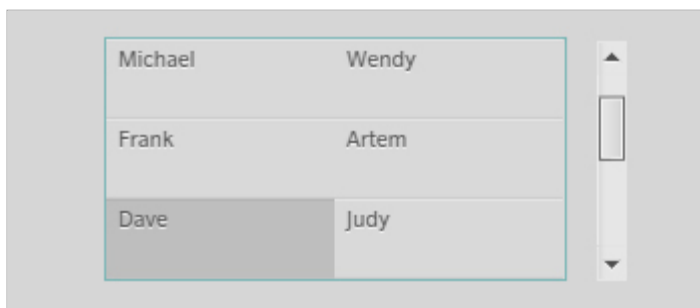
```

//      :
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];

//
//      labelFunction
//      :
list.labelFunction = function(itemObj:Object):String {
//
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];

```

2.4.5 TileList



41: TileList

TileList (gfx.controls.TileList) ScrollingList

ScrollBar

TileList ScrollingList TileList

dataProvider 4 dataProvider

:

```
tileList.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

TileList ListItemRenderer

itemRenderer

ListItemRenderer FLA []

2.4.5.1 ~ 4 h 4 - • •



2.4.5.2 e ¥ • 4 • ¥ z - m u z T u %

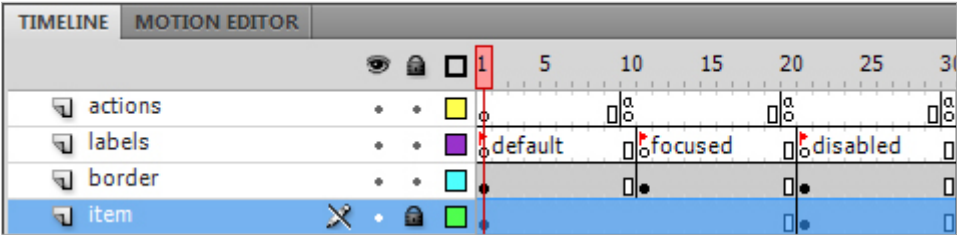
TileList
TileList

2.4.5.3 k x 4 z

TileList focused disabled 3

:

- *default*
- *focused*
- *disabled*



42: TileList

2.4.5.5 V < ¥ z

1 Object

- *type* :
- *target* :

TileList

show	visible	true
hide	visible	false
focusIn		
focusOut		
change	• <i>index</i> : 01	
itemPress	• <i>renderer</i> : CLIK Button • <i>item</i> : dataProviderAS2Object • <i>index</i> : dataProvider01 • <i>mouseIndex</i> : (03)	
itemClick	• <i>renderer</i> : CLIK Button • <i>item</i> : dataProviderAS2Object • <i>index</i> : dataProvider01 • <i>mouseIndex</i> : (03)	

itemDoubleClick

- *renderer* :
Button CLIK
- *item* :
dataProvider AS2 Object
- *index* : dataProvider 0 1
- *mouseIndex* :
(0 3
)

itemRollOver

- *renderer* :
Button CLIK
- *item* :
dataProvider
- *index* : dataProvider 0 1
- *mouseIndex* :
(0 3
)

itemRollOut

- *renderer* :
CLIK Button
- *item* :
dataProvider AS2 Object
- *index* : dataProvider 0 1
- *mouseIndex* :
(0 3
)

TileList

:

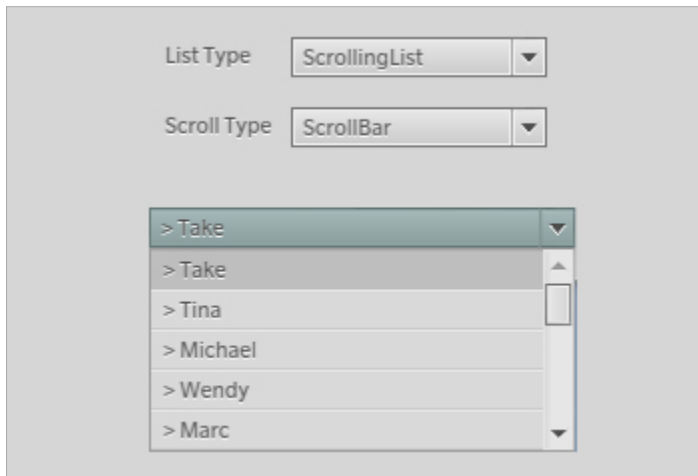
```
myList.addEventListener("focusIn", this, "onListFocused");
function onListFocused(event:Object) {
    trace("tile list was focused!");
    //
}
```

2.4.5.6 „¥z'ev

1 :

```
//  
// TileList      'label'      :  
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];  
  
//      label  
//  
//      :  
list.labelField = "name";  
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];  
  
//  
//      labelFunction  
//      :  
list.labelFunction = function(itemObj:Object):String {  
//  
}  
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];
```

2.4.6 DropdownMenu



43: DropdownMenu

DropdownMenu (gfx.controls.DropdownMenu)

DropdownMenu

ScrollBar

ScrollIndicator

ScrollingList

TileList

dataProvider

DropdownMenu

dataProvider

dataProvider

:

```
dropdownMenu.dataProvider = ["item1", "item2", "item3", "item4"];
```

DropdownMenu

ScrollingList

dropdown

ScrollingList ListItemRenderer

FLA

[

]

DropdownMenu

ScrollBar

ScrollIndicator

DropdownMenu

2.4.6.1 ~ 4 h 4 - • •

DropDownMenu

Enter

ScrollingList TileList ScrollBar

DropDownMenu

DropDownMenu

2.4.6.2 e ¥ • 4 • ¥ z - m u z T u %

DropDownMenu

DropDownMenu

Button

MovieClip

:

- *textField* : () TextField
 - *focusIndicator* : () MovieClip
- MovieClip "show"
- "hide" 2

2.4.6.3 k x 4 z

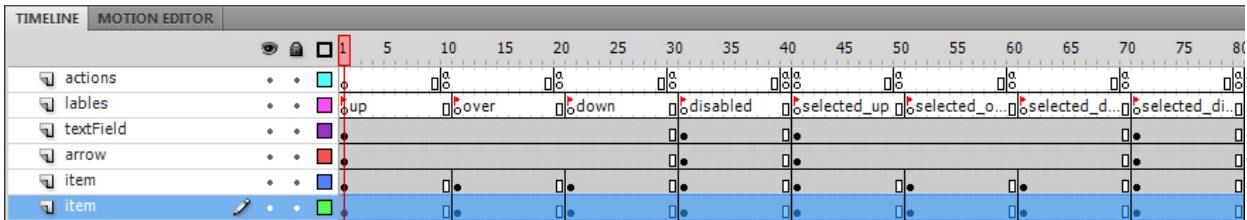
DropDownMenu

ToggleButton

CheckBox

:

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*



44: DropdownMenu

DropdownMenu

Button
DropdownMenu

[Getting Started with CLIK Buttons](#)

2.4.6.4 ^ ' í š) % º Ÿ f x U

DropdownMenu

:

visible	false
disabled	true
dropdown	DropdownMenu (ScrollingList TileList)
dropdownWidth	-1
itemRenderer	DropdownMenu
scrollBar	
margin	
paddingTop	
paddingBottom	
paddingLeft	
paddingRight	
thumbOffsetTop	
thumbOffsetBottom	

thumbSizeFactor	1.0		
offsetX			
offsetY			
extent	offsetX dropdownWidth	-1	
direction		"up"	"down"

2.4.6.5 V < ¥ z

1 Object

- *type* :
- *target* :

DropdownMenu *change*
Button

show	visible	true
hide	visible	false
focusIn		
focusOut		
change	• <i>index</i> : 0 1 • <i>data</i> : AS2 Object	
select	selected	
	• <i>selected</i> : Button selected	
stateChange	• <i>state</i> : Button "up" "over" "down"	

[Getting Started with CLIK Buttons](#)

rollOver	• <i>mouseIndex</i> :	(	0	3	)
rollOut	• <i>mouseIndex</i> :	(	0	3	)
press	• <i>mouseIndex</i> :	(	0	3	)
doubleClick	• <i>mouseIndex</i> :	(	0	3	)
	• <i>doubleClickEnabled</i> :	(	0	3	)
click	• <i>mouseIndex</i> :	(	0	3	)
dragOver	• <i>mouseIndex</i> :	(	0	3	)
dragOut	• <i>mouseIndex</i> :	(	0	3	)
releaseOutside	• <i>mouseIndex</i> :	(	0	3	)

2.4.6.6 „¥ z ' e v

:

```
dropdown.addEventListener("click", this, "onClick");
function onClick(e:Object) {
    if (e.target.isOpen) {
```

```

    //
  }
}

```

DropDownMenu :

```

//          ID          ;
//          ID          /
// fla
// DropDownMenu ScrollingList ScrollBar
attachMovie("DropDownMenu", "dd", this.getNextHighestDepth(),
            {dropdown: "ScrollingList"});
dd.dataProvider = ["one", "two", "three", "four", "five", "six"];

//
//
//
//
//          :
onEnterFrame = function() {
    dd.dropdown.scrollBar = "ScrollBar";
    onEnterFrame = null;
}

```

1 :

```

//
// DropDownMenu          'label'          :
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

//          label
//
//          :
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];

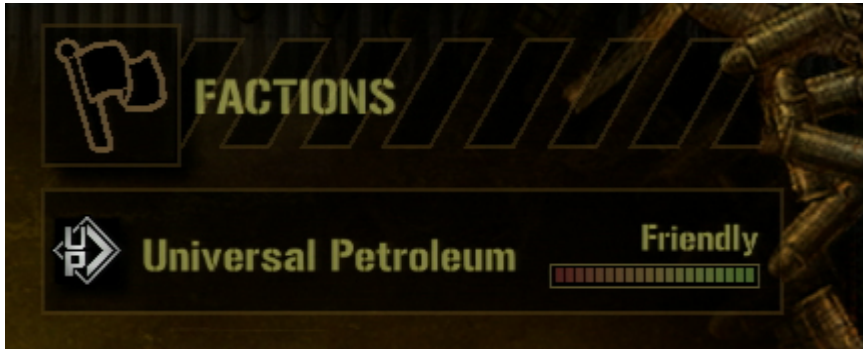
//
//          labelFunction          :
list.labelFunction = function(itemObj:Object):String {
    //
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];

```

2.5 %o Ÿ b ž k q V %o

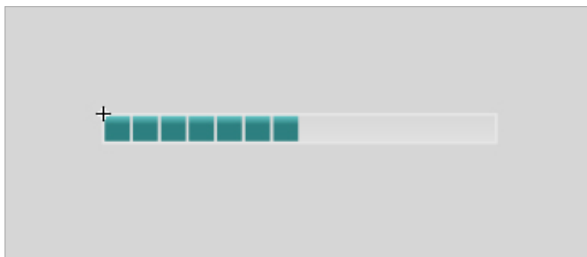
Scaleform CLIK
ProgressBar 2
StatusIndicator

()
StatusIndicator
StatusIndicator
ProgressBar



45: *Mercenaries 2* - FACTIONS

2.5.1 StatusIndicator



46: StatusIndicator

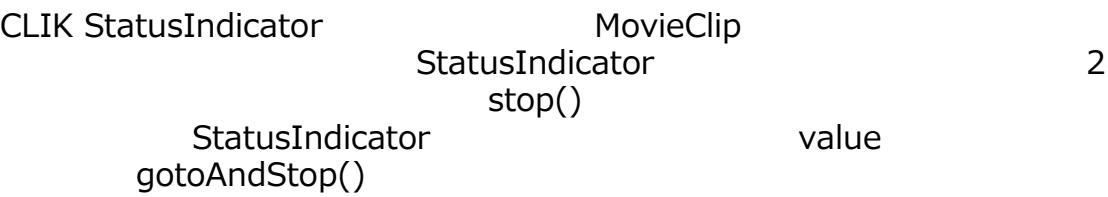
StatusIndicator (gfx.controls.StatusIndicator)

StatusIndicator

2.5.1.1 ~ 4 h 4 - ••

StatusIndicator

2.5.1.2 e ¥ • 4 • ¥ z - m u z T u %



2.5.1.3 k x 4 z

StatusIndicator

2.5.1.4 ^ ' í š) % ¨ Ÿ f x U



visible	false
disabled	true
Value	
minimum	
maximum	

2.5.1.5 V < ¥ z

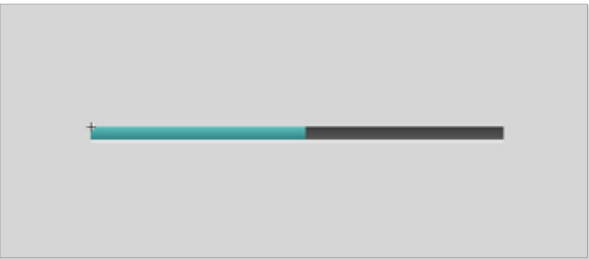
1 Object

- type :
- target :

StatusIndicator

Show	visible	true
Hide	visible	false

2.5.2 ProgressBar



47: ProgressBar

ProgressBar (gfx.controls.ProgressBar)
StatusIndicator

ProgressBar
(bytesLoaded bytesTotal)

2.5.2.1 ~ 4 h 4 - ••

ProgressBar

2.5.2.2 e ¥ • 4 • ¥ z - m u z T u %o

StatusIndicator CLIK ProgressBar MovieClip
ProgressBar
stop() value
2
ProgressBar
gotoAndStop()

2.5.2.3 k x 4 z

ProgressBar

2.5.2.4 ^ ' í š) %o Ÿ f x U

ProgressBar
: MovieClip

visible	false
---------	-------

disabled	true			
Target	bytesLoaded	bytesTotal		ProgressBar
Mode	ProgressBar		"manual"	
	setProgress			
	"polled"		bytesLoaded	bytesTotal
		"event"		
	"progress"	bytesLoaded		bytesTotal

2.5.2.5 V < ¥ z

1 Object

- *type* :
- *target* :

ProgressBar

Show	visible	true
Hide	visible	false
progress	ProgressBar	
complete	ProgressBar	

:

```

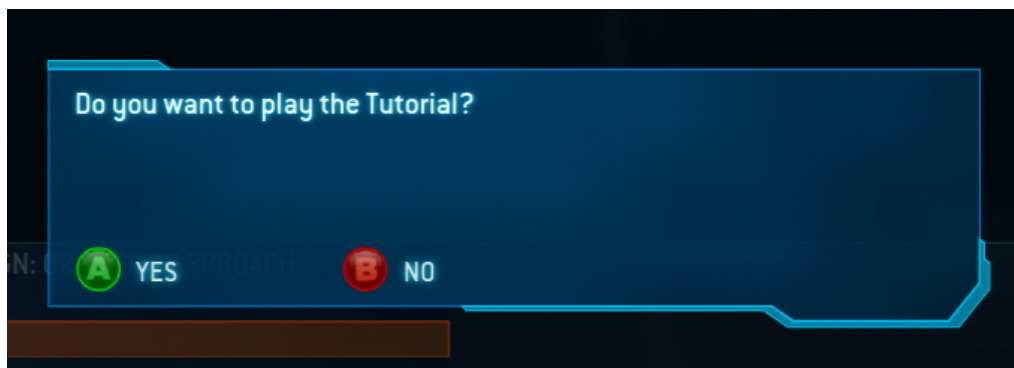
myProgress.addEventListener("progress", this, "onProgress");
myProgress.addEventListener("complete", this, "onProgress");
function onProgress(event:Object) {
    if (event.type == "progress") {
        //
    } else
    {
        trace("Loading complete!");
    }
}

```


2.6 - N - q V %

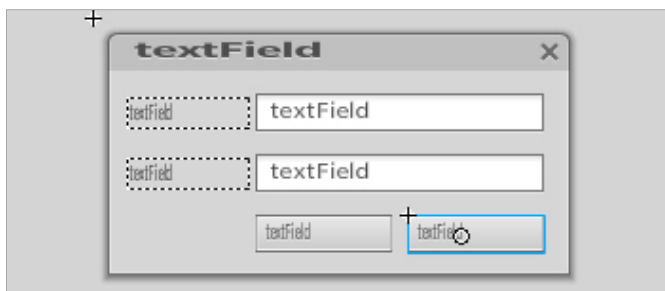
Scaleform CLIK

Dialog UI
 Dialog UILoader ViewStack Dialog
 UILoader
 ViewStack



48: Halo Wars -

2.6.1 Dialog



49: Dialog

CLIK Dialog (gfx.controls.Dialog)

MovieClip
 1 MovieClip () Dialog.show()

2.6.1.1 ~ 4 h 4 - • •

Dialog

2.6.1.2 e ¥ • 4 • ¥ z - m u z T u %

CLIK Dialog MovieClip
:

- *closeBtn* : () CLIK Button
- *cancelBtn* : () CLIK Button
- *submitBtn* : () CLIK Button OK
- *dragBar* : () MovieClip

2.6.1.3 k x 4 z

Dialog MovieClip

2.6.1.4 ^ ' í Š) % ÿ f x U

Dialog MovieClip
:

visible	false
disabled	true

2.6.1.5 V < ¥ z

1 Object

- *type* :
- *target* :

Dialog

Show	visible	true
Hide	visible	false
Submit	Dialog • <i>data</i> : Dialog getSubmitData	Dialog true () getSubmitData AS2 Object
Close	Dialog	

:

```
myDialog.addEventListener("submit", this, "onLoginSubmit");
function onLoginSubmit(event:Object) {
    trace("Received data from dialog: " + event.data);
}
```

2.6.2 UI Loader

CLIK UI Loader (gfx.controls.UI Loader)

SWF/GFX

Scaleform

2.6.2.1 ~ 4 h 4 - • •

UI Loader

SWF/GFX UI Loader

2.6.2.2 e ¥ • 4 • ¥ z - m u z T u %

CLIK UI Loader

MovieClip

:

- *bg* : () MovieClip UI Loader

2.6.2.3 k x 4 z

UILoader SWF/GFX UILoader

2.6.2.4 ^ ' í Š) % º ÿ f x U

```

UILoader
:
MovieClip

```

visible	false	
autoSize	true	UILoader
maintainAspectRatio	true	UILoader false
Source		UILoader SWF/GFX
timeout		
	UILoader	'ioError'

2.6.2.5 $V < \mathbb{Y} Z$

1 Object

- *type* :
- *target* :

UILoader

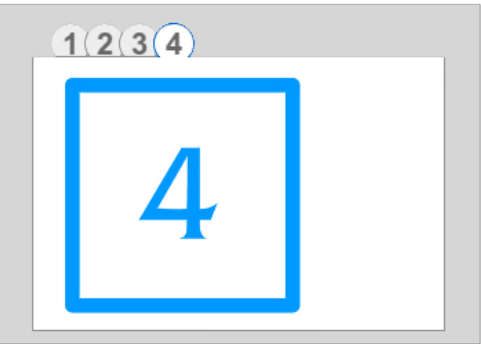
Show	visible	true
Hide	visible	false
progress	a)	b)

Loaded	0	100
complete		
ioError	source	

:

```
myUILoader.addEventListener("ioError", this, "onLoadingError");
function onLoadingError(event:Object) {
    displayErrorMessage("Could not load" + event.target.source);
}
```

2.6.3 ViewStack



50: ViewStack

```
CLIK ViewStack (gfx.controls.ViewStack)
1
Accordion
demos
RadioButton
)
TabBox
( 2
ViewStack
```

2.6.3.1 ~ 4 h 4 - ••

ViewStack ViewStack

2.6.3.2 e ¥ • 4 • ¥ z - m u z T u %

CLIK ViewStack MovieClip

2.6.3.3 k x 4 z

ViewStack ViewStack

2.6.3.4 ^ ' í š) % Ÿ f x U

ViewStack : MovieClip

visible	false
Cache	true
	ViewStack targetGroup ()
targetGroup	ButtonGroup 'change'
	RadioButton ID ID Flash IDE [

2.6.3.5 V < ¥ z

ViewStack

3 T 4 z - •

Scaleform CLIK

3.1 < k z %> a x U k

Scaleform CLIK

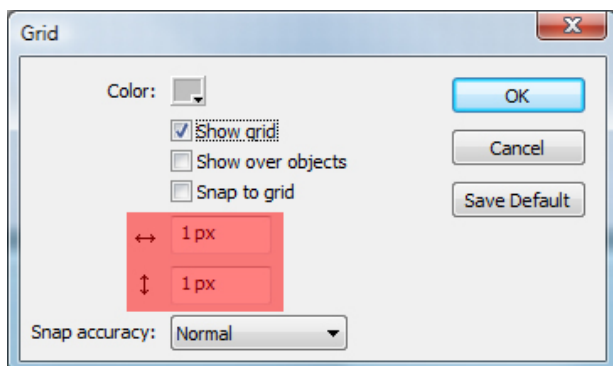
3.1.1 † a m • f 4 ‡ Y a z) V “ 4 j

CLIK

Flash

:

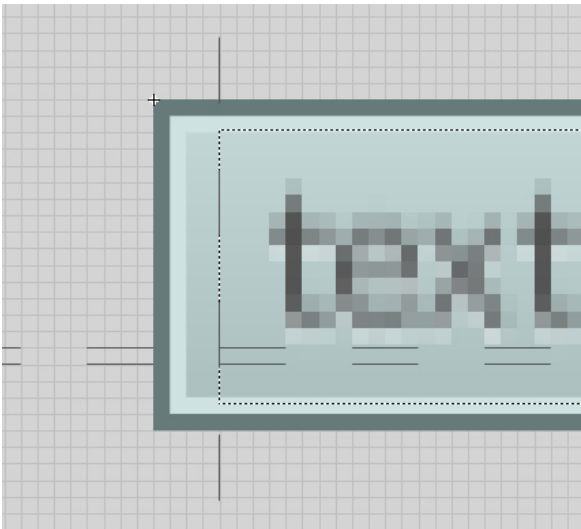
1. Flash []
2. [] []
3. Flash []
4. [] []
5. 1 px
6. [OK]



51: []

SWF

: 2 2



52:

3.1.2 • k a

Flash

Scaleform

Flash

PNG Photoshop®

3.1.3 T } “ 4 i ™ ¥

1

()

3.1.4 ž V – 4 ' \ ° ‰ œ ‘ x U ^

Flash

1

3.1.5 3 Ä) k _ ¥

()

3.1.6 2 - 8 -

2

2

:

- 16x16
- 32x32
- 64x64
- 128x128
- 128x256
- 256x256
- 512x256
- 512x512

3.2 C t - ; ' Ó ĭ 4 a ‡ Ÿ 4

Flash

Scaleform CLIK

3.2.1 e ¥ • 4 • ¥ z - 3 2

Flash

2

3.2.1.1 e ¥ • 4 • ¥ z - 3 2 (í ç 1)

Button () CLIK
FLA

FLA
:

1. *CLIK_Components fla*

2. []

(Button)
FLA [

3. [] []

4. FLA ['Button']

5. FLA []

6. []

7. []

8. []

3.2.1.2 e ¥ • 4 • ¥ z - 3 2 (í ç 2)

()

:

1. [] []

2. [] []
(gfx.controlsButton) Ctrl+C

3. []

4. []
5. [] [ActionScript []]

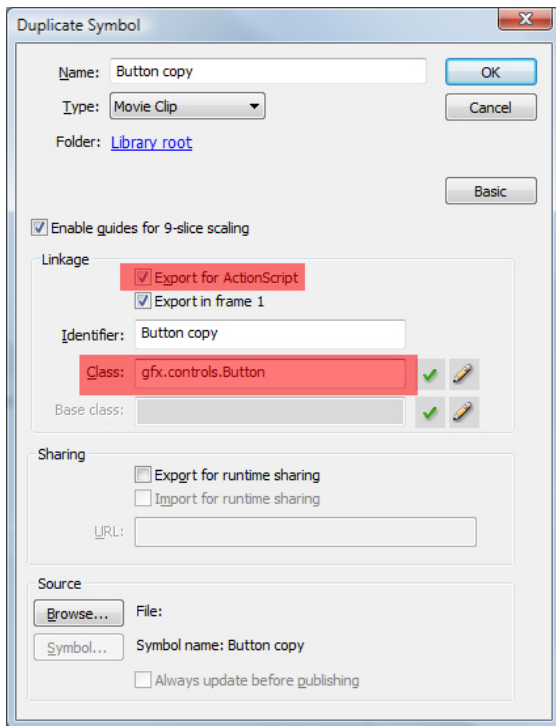
6. [] Ctrl+V

7. [OK]

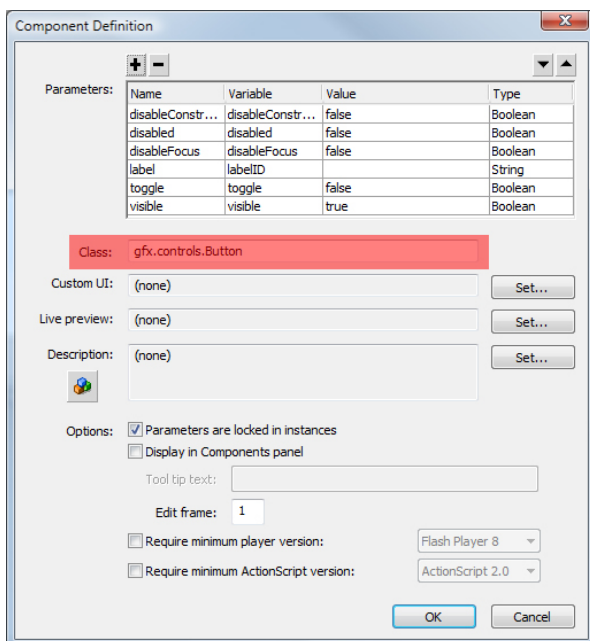
8. []
 9. []

Ctrl+V

10.[OK]



53: ()



54: ([1])

3.3 k_} ¥ b - {

Scaleform CLIK

1

CLIK

CLIK
FLA []

Flash

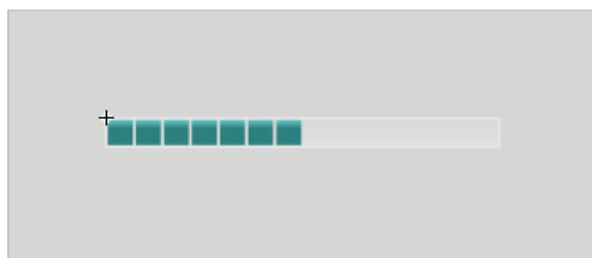
:

- Flash
- Flash
- Photoshop Illustrator®

Flash

[Getting Started with CLIK](#)

3.3.1 StatusIndicator - k_} ¥ b



55: StatusIndicator

StatusIndicator

Button

StatusIndicator

indicator track

2

track

indicator

1 Photoshop

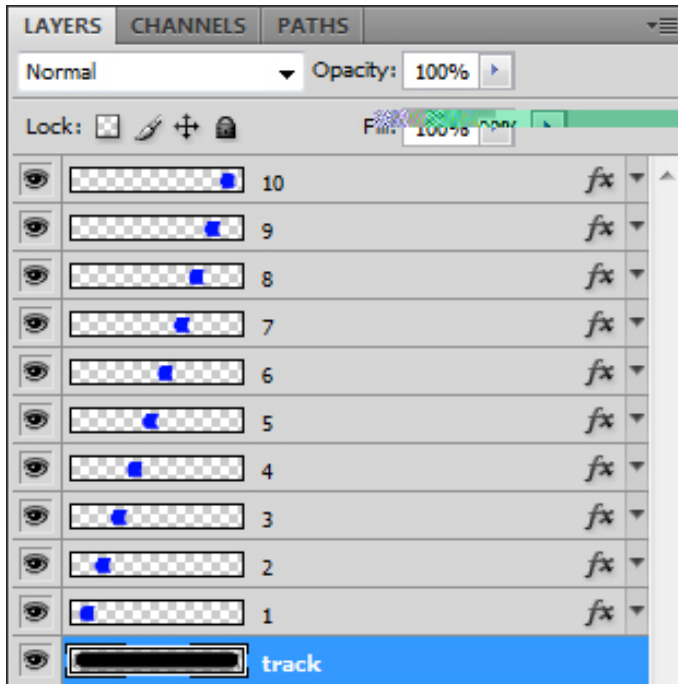
StatusIndicator

1

StatusIndicator



56: StatusIndicator PSD



57: StatusIndicator PSD Photoshop

1. StatusIndicator Photoshop
2. PSD
3. Flash [] []->[]
4. StatusIndicator PSD
5. [] [] [Flash]
6. [OK]
7. PSD Flash

10

1

(

10

Flash

1

10

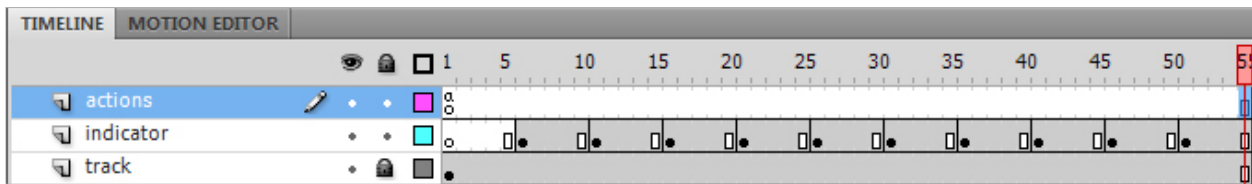
)

[

1

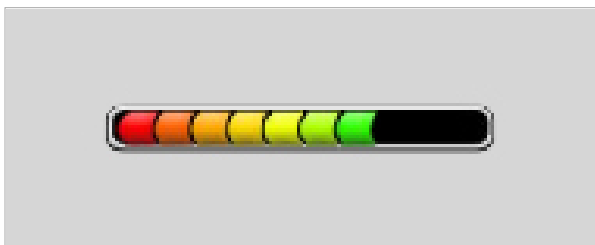
10
- 8.
- 9.
10. [1] 6

11.[1] 11 11
 12.[2] Ctrl+X
 13.[1] 11 Ctrl+Shift+V
 14.10 [1] (1) 5
 11 [3] 16 2] [4]
 21 10
 PSD 51
 15. (2-10)
 16. 5 [55
 17. indicator
 18. track track
 19.[1] 'indicator'
 20. indicator track actions
 track indicator



58: ()

21.StatusIndicator
 22. 1 10
 23.
 24.



59: StatusIndicator

3.4 ¢ [¥ z ' ¨ 4] > V I

Scaleform CLIK

3.4.1 . @ *

Scaleform () SWF
Scaleform
Flash UI

3.4.2 ¢ [¥ z - ¢ @ b >

Flash Player Scaleform
Scaleform Player
Scaleform Slate Mobile textField
Slate Mobile (CJK)
Slate Mobile CJK ASCII
textField Scaleform (3.4.4
Scaleform Scaleform
UI

3.4.3 textField * ¢ [¥ z Q ¢ @ b ?

textField (textField
textField] ([]->[
[OK] FLA textField FLA

3.4.4 Scaleform - 4] > V I i k x '

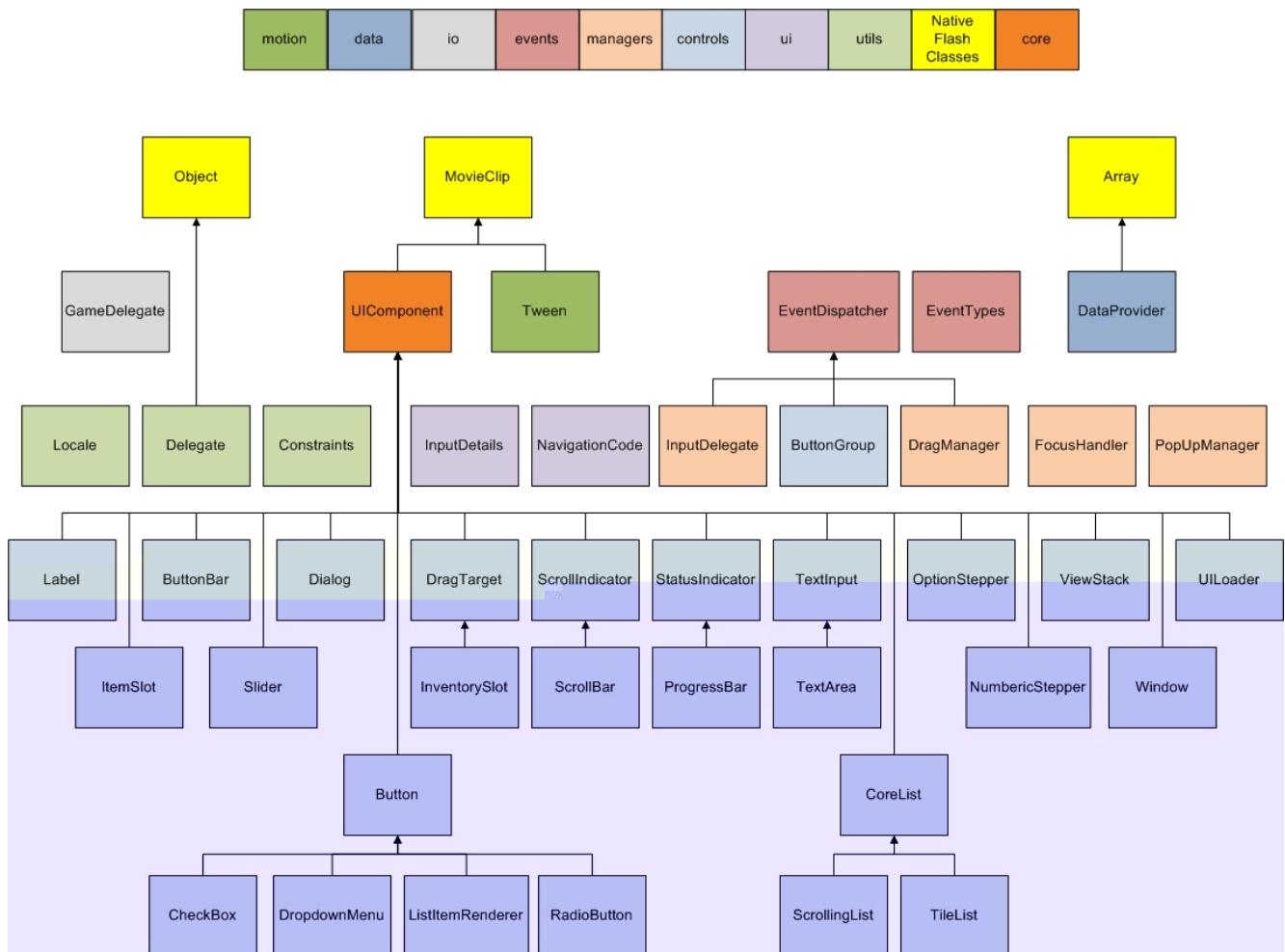
Scaleform
/
SWF
SWF
Scaleform
textField
textField
gfxfontlib fla
Scaleform
textField
Scaleform
(
) Scaleform
fontconfig txt
textField
fontconfig txt
Scaleform
textField '\$TITLE'
\$TITLE=Scaleform
textField 'Scaleform'
Scaleform
Scaleform

[Font Overview](#)

4 %o Ÿ b > ‘ ¥ b - •

Scaleform

CLIK



60: Scaleform CLIK

4.1 UIComponent

Scaleform CLIK

UIComponent (gfx.core.UIComponent)
 UIComponent CLIK UIComponent Scaleform

UIComponent Flash 8 MovieClip
 Flash MovieClip / UIComponent :

- *disabled;*
 - *visible;*
 - *focused;*
 - *width;*
 - *height;*
 - *displayFocus :* true
- focused
-

UIComponent :

- *configUI :*
 - *draw :*
 - *changeFocus :*
 - *scrollWheel :*
-

UIComponent Mix-in EventDispatcher

4.1.1 B 0

onLoad() :

- MovieClip

- configUI()
- validateNow()
- draw()

4.2 e ¥ • 4 • ¥ z k x 4 z

Scaleform CLIK

3

4.2.1 Ž q ¥ e ¥ • 4 • ¥ z

CheckBox Button
 ScrollBar
 (gfx.controls.Button)

ListItemRenderer CLIK
 RadioButton
 Button

:

- *up*
- *over*
- *down*
- *disabled*

"selected_"

CLIK Button

selected

:

- *up*
- *over*
- *down*
- *disabled*

- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

: CLIK

[Getting Started with CLIK](#)

[Buttons](#)

CLIK Button getStatePrefixes()

:

```
private function getStatePrefixes():Array {
    return (_selected) ? ["selected_", ""] : [""];
}
```

CLIK
getStatePrefixes() "selected_"
selected

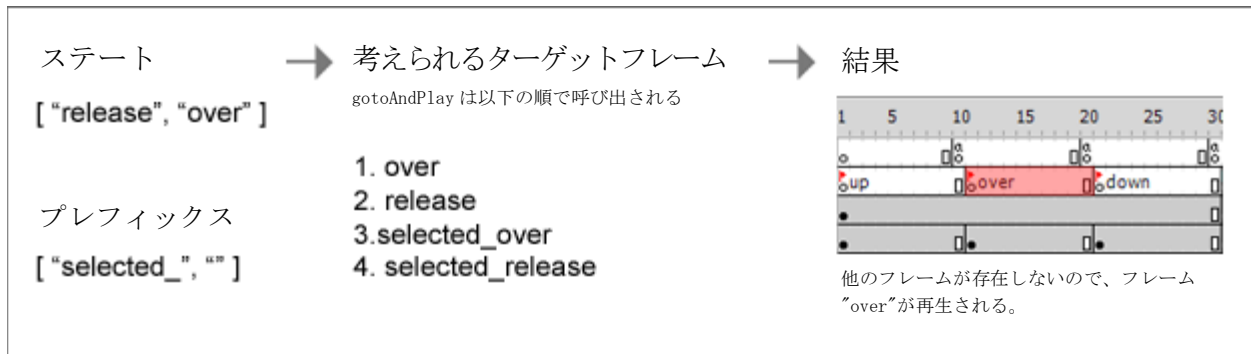
Button stateMap
CLIK Button

:

```
private var stateMap:Object = {
    up: ["up"],
    over: ["over"],
    down: ["down"],
    release: ["release", "over"],
    out: ["out", "up"],
    disabled: ["disabled"],
    selecting: ["selecting", "over"],
    kb_selecting: ["kb_selecting", "up"],
    kb_release: ["kb_release", "out", "up"],
    kb_down: ["kb_down", "down"]
}
```

getStatePrefixes()

:



61:

4.2.2 $\tilde{Z} q \neq 0 - V \neq q > a \times U^{\wedge} \quad e \neq \bullet 4 \bullet \neq z$

CLIK

ScrollingList OptionStepper Slider TextArea

:

- *default*
- *focused*
- *disabled*

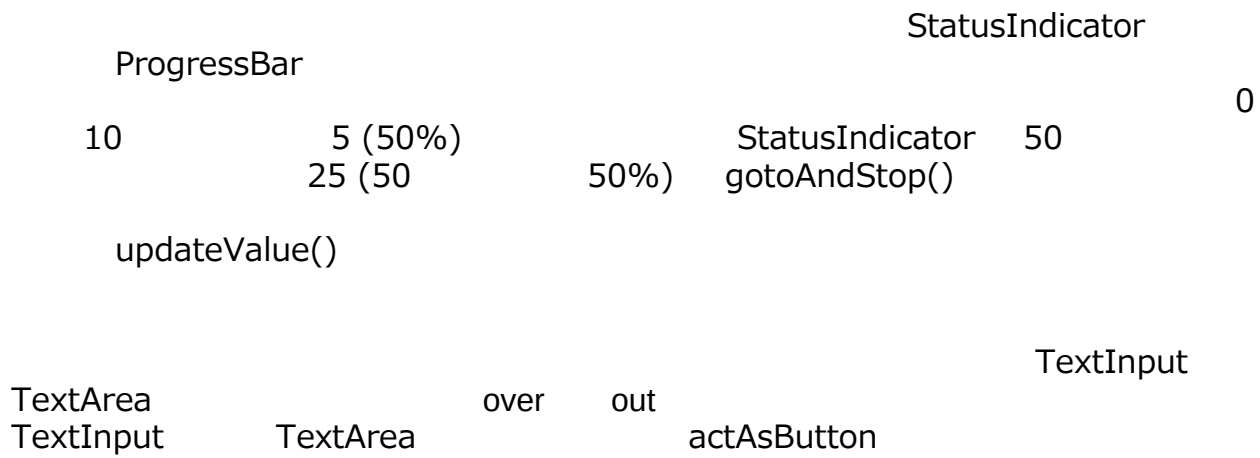
4.2.3 $6 V \neq q > a \times U^{\wedge} \quad e \neq \bullet 4 \bullet \neq z$

Label

:

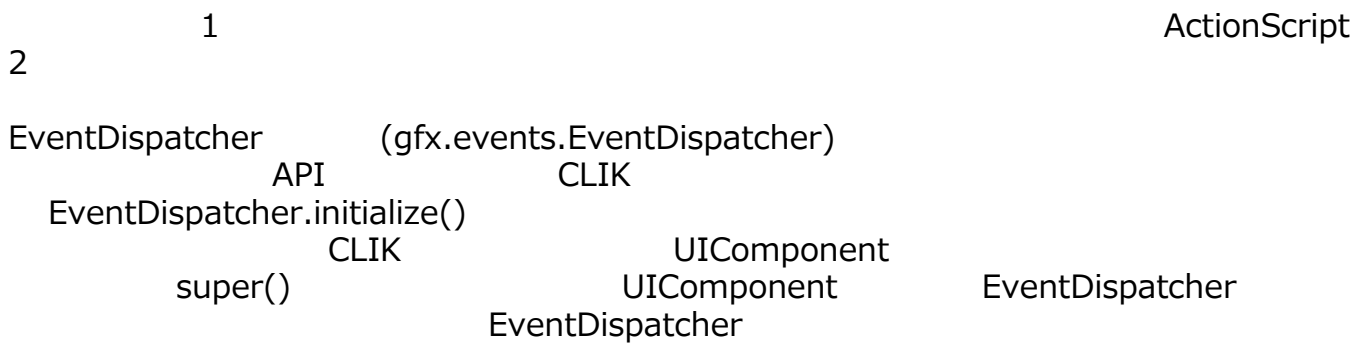
- *default*
- *disabled*

4.2.4 Í ð) c 4 k



4.3 V < ¥ z " y •

Scaleform CLIK



4.3.1 ž ÷ ç ' < k z %o > a x U k

4.3.1.1 V < ¥ z - g ^ k a > V ^



callBack

callBack

4.4.1 Ž ÷ ç ' < k z %o > a x U k

CLIK

```

Player
(
    Tab
    focused
    true
)
:

```

```
Selection.setFocus(firstComponentOrMovieClip);
```

MovieClip (onRelease onRollOver) Scaleform Flash
Player

MovieClip focusEnabled tabEnabled

ScrollBar
focusTarget

focusTarget FocusHandler focusTarget

Player

displayFocus true

Slider

Slider
UIComponent.changeFocus() focused

```

function changeFocus():Void {
    track.displayFocus = _focused;
}

```

tabEnabled false

```
background.tabEnabled = false;
```


4.4.2

ScrollingList OptionStepper ButtonBar

Flash Scaleform Selection

:

1. (OptionStepper
tabEnabled = false
2. focusTarget
focusEnabled = true
3. changeFocus displayFocus
true

4.5

Scaleform CLIK

Flash 8 MovieClip
MovieClip

CLIK

4.5.1

InputDelegate
(gfx.managers.InputDelegate) InputDelegate

InputDetails (gfx.ui.InputDetails)
InputDelegate.readInput()
InputDetails

InputDelegate

InputDetails :

-
-
-

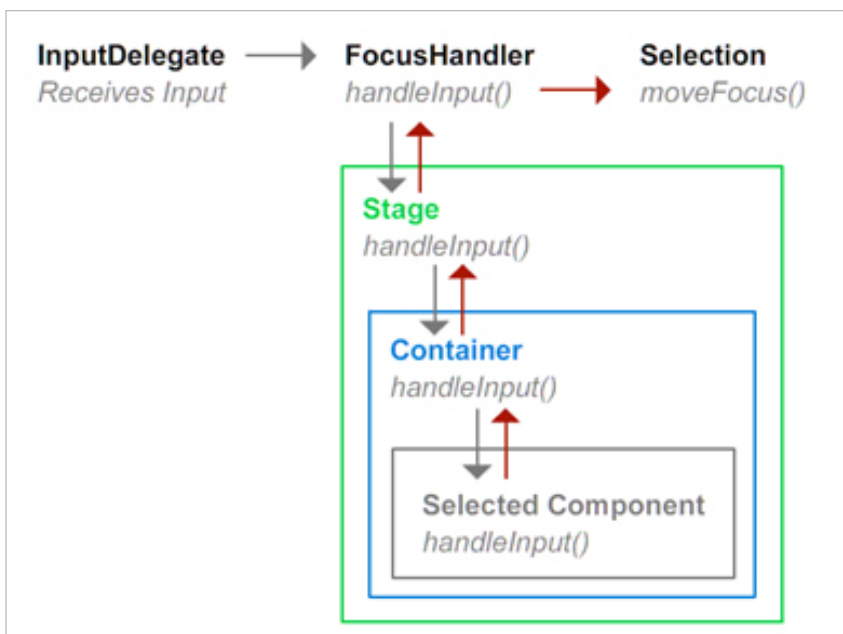
key up key down
 InputDetails
 “keyUp” “keyDown”
 • navEquivalent ()
 NavigationCode (gfx.ui.NavigationCode)

FocusHandler InputDelegate

handleInput()

InputDetails

pathToFocus



62: handleInput()

FocusHandler handleInput()

false

Scaleform Player

null

navEquivalent

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
```

```

        if (details.navEquivalent == "left")
        {
            //      NavigationCode.LEFT
            doSomething();
            return true;
        }
        return false;
    }
}

```

pathToFocus
 true false
 handleInput

```

function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
    var nextItem:MovieClip = pathToFocus.shift();
    var handled:Boolean = nextItem.handleInput(details, pathToFocus);
    if (handled) { return true; }
    //
    return false; //      true
}

```

pathToFocus DropdownMenu handleInput

```

function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
    var handled:Boolean = dropdown.handleInput(details);
    if (handled) { return true; }
    //
    return false; //      true
}

```

InputDelegate.instance

FocusHandler

```

InputDelegate.instance.addEventListener("input", this, "handleInput");
function handleInput(event:Object):Void {
    var details:InputDetails = event.details;
    if (details.value == Key.TAB) { doSomething(); }
}

```

InputDelegate
 InputDelegate

W A S D

4.5.2 3 / - • X k] 4 o •

Wii

1 SWF CLIK

mouseIndex

```
myButton.addEventListener("click", this, "onCursorClick");
function onCursorClick(event:Object):Void {
    switch (event.mouseIndex) {
        ...
    }
}
```

4.6 î • ò

```
btnInstance.setSize(100,22);
btnInstance.width = 200;
btnInstance.label = "text";
btnInstance.toggle = true;
btnInstance.selected = true;
btnInstance.data = obj;
```



The diagram shows a sequence of six property assignments for `btnInstance`: `setSize(100,22)`, `width = 200`, `label = "text"`, `toggle = true`, `selected = true`, and `data = obj`. These are grouped by a bracket on the right, which points to an arrow leading to `invalidate()`, which in turn points to `draw()`.

63:

CLIK

4.6.1 ž ÷ ç ' < k z %o > a x U k

UIComponent.invalidate() (setter)
UIComponent.draw() invalidate()

draw()

UIComponent.validateNow()

4.7 e ¥ • 4 • ¥ z - k c 4 œ ¥ b

Scaleform CLIK 2 :

1.

2.

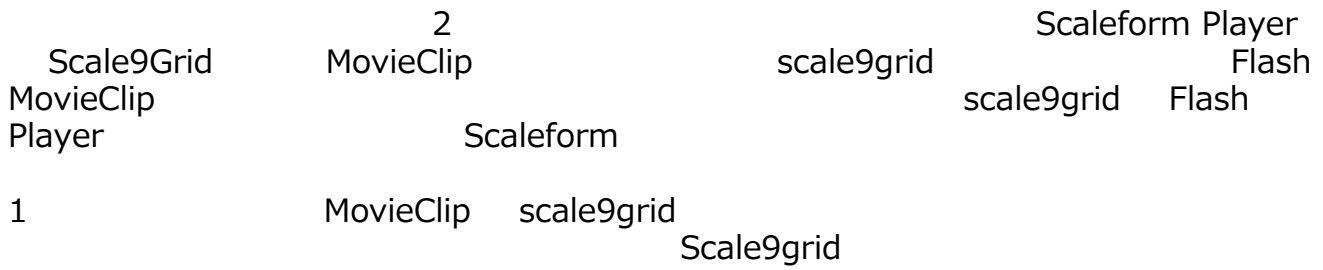
scale9grid

Flash Flash 9 Flex “

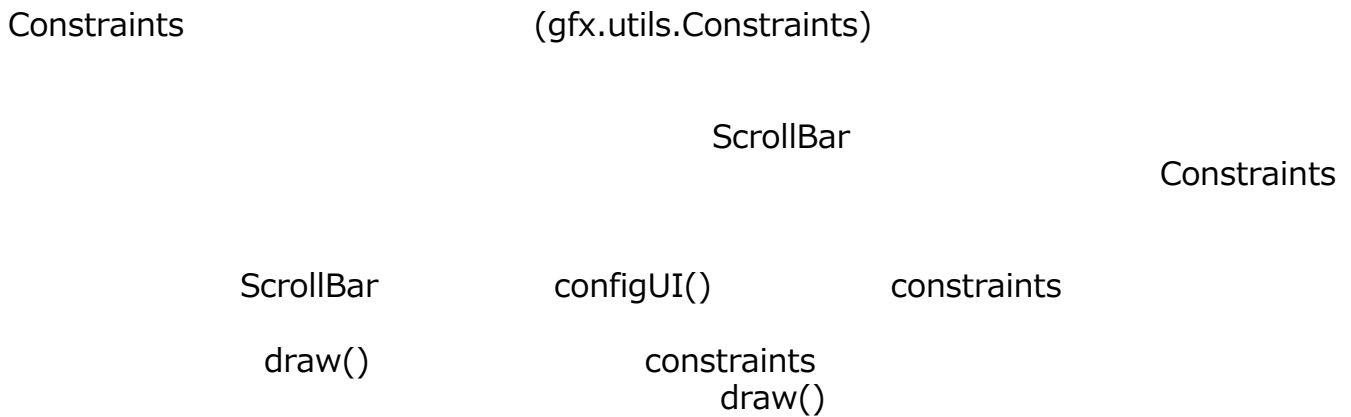
Scaleform 9
CLIK
CLIK

Flash IDE width height
setSize()
Flash IDE
LivePreview MovieClip
Scaleform Player
CLIK Launcher

4.7.1 Scale9Grid



4.7.2 Constraints



```

private function configUI():Void {
    ...
    constraints = new Constraints(this);
    // upArrow
    constraints.addElements(downArrow, Constraints.BOTTOM);
    constraints.addElenent(track, Constraints.TOP|Constraints.BOTTOM);
    ...
}

private function draw():Void {
    ...
    constraints.update(__width, __height);
    ...
}
  
```

4.8 e ¥ • 4 • ¥ z ' y 4 q - m u z

dataProvider
(gfx.interfaces.IDataProvider)

dataProvider
Scaleform CLIK IDataProvider
API

dataProvider

dataProvider

dataProvider
OptionStepper
API

DropDownMenu
IDataProvider

CoreList (ScrollingList
IDataProvider

TileList)
CLIK

4.8.1 ž ÷ ç ' < k z %o > a x U k

initialize()
dataProvider

DataProvider
dataProvider

(gfx.data.DataProvider)
ActionScript
dataProvider

IDataProvider

dataProvider :

```
myComponent.dataProvider = [ "data1",
                              4.3,
                              {label:"anObjectElement", value:6} ];
```

dataProvider :

- *length* : getter
 - *requestItemAt* : dataProvider
1 OptionStepper
 - *requestItemRange* : dataProvider
ScrollingList
 - *indexOf* :
 - *invalidate* : dataProvider
"dataChange"
- dataProvider

length

dataProvider

ActionScript 2

dataProvider

ExternalInterface.call

4.9 r V | ' u a T } " 4 i ™ ¥

Scaleform CLIK
Scaleform
Tween

Flash 8 Tween
Tween
mx.transitions.easing.*

(gfx.motion.Tween)

CLIK Tween

Flash
Flash 8 MovieClip
CLIK

CLIK Tween onEnterFrame

MovieClip
MovieClip

1

MovieClip

MovieClip

4.9.1 ž ÷ ç ' < k z %o > a x U k

Tween MovieClip 2 tweenTo() tweenFrom()
tweenTo() MovieClip
tweenFrom()

MovieClip
Tween.init()

:

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init(); // MovieClip
// 1
myMovieClip.tweenTo(1, {_x: 200, _alpha: 0}, Strong.easeIn);
```


onTweenComplete()

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init();
mc.onTweenComplete = function() {
    trace("Tween has finished!");
}
mc.TweenTo(5, {_rotation: 20}, Bounce.easeOut);
```

4.10 • u % T u % - g • 4 z

Scaleform CLIK PopUpManager (gfx.managers.PopUpManager)
Dialog
PopUpManager

4.10.1 ž ÷ ç ' < k z % > a x U k

PopUpManager
createPopUp()
) ID MovieClip (CLIK context
relativeTo
MovieClip

```
import gfx.managers.PopUpManager;
PopUpManager.createPopUp(context, "MyToolTipLinkageID",
    {_x: 200, _y:200}, relativeTo);
```

Scaleform topmostLevel

SWF

topmostLevel

topmostLevel
Z

topmostLevel true

4.11 { > u b m { Ÿ u %

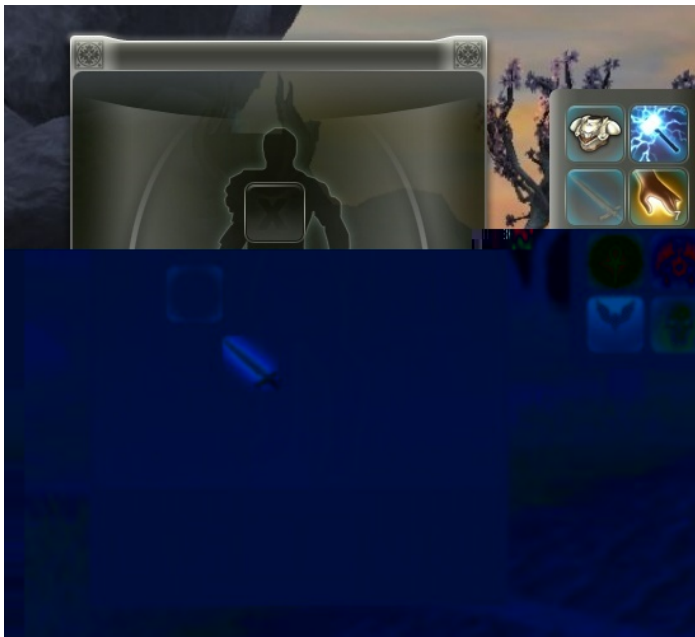
DragManager (gfx.managers.DragManager)

/

startDrag() MovieClip ID

stopDrag()
DragManager InputDelegate

4.11.1 ž ÷ ç ' < k z % > a x U k



64:

DragManager DragManager

Scaleform CLIK DragTarget (gfx.controls.DragTarget)
DragTarget UIComponent CLIK
DragManager

```

DragTarget
/
DragTarget
DragTarget
dragBegin dragEnd
DragTarget DragManager
Inventory Demo DragTarget
DragManager DragTarget &
2 InventorySlot ItemSlot
2 CLIK
InventorySlot (gfx.controls.InventorySlot)
DragTarget DragManager
InventorySlot
ItemSlot (gfx.controls.ItemSlot) InventorySlot
ItemSlot DragTarget 2 CLIK Button
& ItemSlot

```

4.12 - N

4.12.1 Delegate

```

Delegate (gfx.utils.Delegate)
Scaleform CLIK
DragManager :

```

```

eventProvider.onMouseMove = Delegate.create(this, doDrag);

```

4.12.2 Locale

```

Locale (gfx.utils.Locale)
Scaleform
ActionScript

```

Button Label TextInput
Locale.getTranslatedString()

Locale

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Scaleform CLIK

5.1 f Š c

Scaleform CLIK

5.1.1 2 # - textField Q ß ListItem Renderer



65: 2

ScrollingList

```

ScrollingList
    ListItemRenderer 1
        textField
    textField
    ListItemRenderer 2
        textField
    ListItemRenderer 2
        textField
    ListItemRenderer
    ScrollingList
    dataProvider
    :

```

```

list.dataProvider = [{fname: "Michael", lname: "Jordan"},
                    {fname: "Roger",    lname: "Federer"},
                    {fname: "Michael",  lname: "Schumacher"},
                    {fname: "Tiger",    lname: "Woods"},
                    {fname: "Babe",     lname: "Ruth"},
                    {fname: "Wayne",    lname: "Gretzky"},
                    {fname: "Usain",    lname: "Bolt"}];

```

```

dataProvider      2      2      fname      lname
                2      2      textField
                ListItemRenderer      1      textField      2
textField
ListItemRenderer      MyItemRenderer
                ListItemRenderer      2
textField      (      FLA
    MyItemRenderer.as      :

```

```

import gfx.controls.ListItemRenderer;

class MyItemRenderer extends ListItemRenderer {

    public var textField1:TextField;
    public var textField2:TextField;

    public function MyItemRenderer() { super(); }

    public function setData(data:Object):Void {
        this.data = data;
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }

    private function updateAfterStateChange():Void {
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }
}

```

```

ListItemRenderer      setData      updateAfterStateChange
                setData
    (ScrollingList      TileList      )
    ListItemRenderer      1      textField
MyItemRenderer      textField      updateAfterStateChange
                ListItemRenderer
                textField

                dataProvider
    ListItemRenderer
                ListItemRenderer

ListItemRenderer      'textField1'      'textField2'      2      textField

```

```

        MyItemRenderer
        itemRenderer
        'MyItemRenderer'

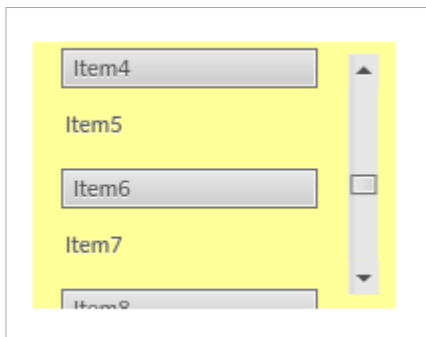
        FLA
        2

        dataProvider

        fname  lname

```

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66: CLIK

CLIK ScrollBar

:

- *actions* : ActionScript
- *scrollbar* : CLIK ScrollBar
'sb'
- *mask* : MovieClip
- *content* :
- *background* :

'content' y 0

code

ActionScript

:

```
// 133 ( )  
sb.setScrollProperties(1, 0, content._height - 133);  
sb.position = 0;  
  
sb.addEventListener("scroll", this, "onScroll");  
function onScroll() {  
    content._y = -sb.position;  
}
```

setScrollProperties

Scaleform

HUD
Scaleform

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Scaleform CLIK

Flash ActionScript 2

ScrollingList TreeView

67: TreeView

TreeView Demo (*Resources/AS2/CLIK/demos/TreeViewDemo.fla*)
ListItemRenderer DataProvider ScrollingList

ScrollingList :

-

-

ListItemRenderer

-

ListItemRenderer

ScrollingList
 TreeViewItemRenderer
 DataProvider
 ListItemRenderer
 TreeViewDataProvider
 TreeViewConstants

Resources/AS2/CLIK/demos/com/scaleform

ActionScript 2

Object
 Object :

```
var root:Object =
{
    label: "System",
    nodes:[
        {label: "Applications",
          nodes:[
              {label: "CGStudio",
                nodes:[
                    {label: "Data"},
                    {label: "Samples"},
                    {label: "Config"}
                ]
              },
              {label: "Money Manager"},
              {label: "Role Call v6"},
              {label: "Super Draft",
                nodes:[
                    {label: "Templates"},
                    ...
                ]
              }
            ]
        }
    ]
}
```

label nodes 2 Object
 Array ScrollingList

DataProvider

TreeViewDataProvider
 DataProvider
 ListItemRenderer

/ /
 TreeViewDataProvider CLIK

TreeViewItemRenderer

ScrollingList

TreeViewDataProvider

ListItemRenderer

TreeViewItemRenderer

ListItemRenderer textField

TreeViewConstants

:

- :
 -
 -
- 2

ScrollingList

ScrollBar

/
ScrollingList

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68: Window

Window Demo (*Resources/AS2/CLIK/demos/WindowDemo.fla*)

Window

Resources/AS2/CLIK/demos/com/scaleform

Window

UIComponent

CLIK

Window

Window

:

-
-
-
-
-

()

Window

:

Title	Window			
formPadding	Window			
formType	'symbol' 'swf'			
	'symbol'			
	'swf'	SWF		
formSource	)	SWF	(formType 'swf	'symbol'
			)	
allowResize	/			
minWidth,	max			
maxWidth,	min			
minHeight,	max			
maxHeight	max 0			
	min			

:

```
[Inspectable(name="formType", enumeration="symbol,swf")
private var _formType:String = "symbol";
```

Inspectable
Inspectable
_formType

Flash
formType

getter setter

```
[Inspectable(defaultValue="Title")]  
public function get title():String { return _title; }  
public function set title(value:String):Void {  
    _title = value;  
    invalidate();  
}
```

Window

CLIK

Button

Window
2

UIComponent
constraints

configUI() configUI() draw()
draw()
constraints

configForm()

Flash 8 MovieClipLoader
Scaleform

MovieClipLoader.onLoadComplete()

configForm()

constraints

Window

draw()

constraints
constraints

validateNow()

UIComponent

draw()
draw()
UIComponent

validateNow()
Window Demo

draw()

validateNow()

SWF

Window

Window

2

```
private function onMouseDown() {  
    var targetObj:Object = Mouse.getTopMostEntity();  
    while (targetObj != null && targetObj != _root)  
    {  
        if (targetObj == this) {  
            swapDepths(_parent.getNextHighestDepth());  
            return;  
        }  
        targetObj = targetObj._parent;  
    }  
}
```

1

WindowManager

WindowManager

Window